

5 Differential Equations (Higher Order)

5.1. Introduction

A differential equation (DE) is a mathematical equation for an unknown function of one or several variables that relates the values of the function itself and its derivatives of various orders. Differential equations play a prominent role in Engineering, physics, economics and other disciplines. An example of modeling a real world problem using differential equations is determination of the velocity of a ball falling through the air, considering only gravity and air resistance. The balls acceleration towards the ground is the acceleration due to gravity, minus the deceleration due to air resistance. Gravity is constant; however, air resistance is proportional to the ball's acceleration, which is the derivatives of its velocity, hence depending upon the velocity. Hence, finding the velocity as a function of time requires solving differential equation.

Ordinary differential equations (ODE) are classified according to the order of the highest derivative with respect to the variable appearing in the equation. The most cases for applications are first order and second order DE's. In the classical literature also distinction is made between DE's explicitly solved with respect to highest derivative and DE's in an implicit form.

5.2. Linear differential equations of higher order-with constant co-efficients

Definition: A differential equation, which contains the derivative or differentials of one or more dependent variables with respect to a single independent variable, is called an ordinary

differential equation. The general form of a linear differential equation of the n^{th} order with constant co-efficient is

$$a_0 \frac{d^n y}{dx^n} + a_1 \frac{d^{n-1} y}{dx^{n-1}} + \dots + a_{n-1} \frac{dy}{dx} + a_n y = X \quad \dots(1)$$

where $a_0 (\neq 0), a_1, a_2, \dots, a_n$ are constants and X is a function of x . If we use the differential

operator symbols $D = \frac{d}{dx}$, $D^2 = \frac{d^2}{dx^2}$, ..., $D^n = \frac{d^n}{dx^n}$, equation (1) becomes

$$(a_0 D^n + a_1 D^{n-1} + \dots + a_{n-1} D + a_n) y = X \quad \dots(2)$$

or $f(D)y = X$, where $f(D)$ is a polynomial in D , which may be treated as an algebraic quantity. When $X = 0$, equation (2) becomes $f(D)y = 0$... (3)

Equation (3) is called the homogeneous equation corresponding to equation (2).

General solution of (2) is $y = u + v$, where $y = u$ is the general solution of (3), that contains n arbitrary constants and $y = v$ is a particular solution of (2), that does not contain any arbitrary constants. u is called the **complementary function** (C.F.) and v is called the **particular integral** (P.I.) of the solution of equation (2).

Complementary function

To find the C.F. of the D.E. (2), we require the general solution of (3). We first write down the auxiliary equation (A.E.) $f(m) = 0$ or $a_0 m^n + a_1 m^{n-1} + \dots + a_n = 0$... (4)

which is obtained by simply replacing D by m in the operator polynomial, and then by equating to zero. The solution of equation (3) depends on the nature of roots of the A.E. (4).

Let the roots of the A.E. be $m_1, m_2, m_3 \dots$

Case (i) : If all the roots of the A.E are real and distinct, then the complementary function is given by $y = c_1 e^{m_1 x} + c_2 e^{m_2 x} + \dots + c_n e^{m_n x}$

Case (ii) : If any two roots are real and equal say $m_1 = m_2 = m$, and the other roots are real and unequal, then the C.F. is given by $y = (c_1 x + c_2) e^{mx} + c_3 e^{m_3 x} + c_4 e^{m_4 x} + \dots + c_n e^{m_n x}$

Case (iii) : If any three roots are real and equal say $m_1 = m_2 = m_3 = m$, and the other roots are real and unequal, then the C.F. is given by $y = (c_1 x^2 + c_2 x + c_3) e^{mx} + c_4 e^{m_4 x} + \dots + c_n e^{m_n x}$

Case(iv): If one pair of roots is complex say $m_1 = \alpha + i\beta$, $m_2 = \alpha - i\beta$ and the other roots are real and unequal then the C.F is given by

$$y = e^{\alpha x} (c_1 \cos \beta x + c_2 \sin \beta x) + c_3 e^{m_3 x} + \dots + c_n e^{m_n x}$$

Case (v): If two pairs of roots are complex, then the C.F is given by

$$y = e^{\alpha x} [(c_1 x + c_2) \cos \beta x + (c_3 x + c_4) \sin \beta x] + c_5 e^{m_5 x} + \dots + c_n e^{m_n x} \quad \text{where } c_1, c_2, \dots, c_n \text{ are arbitrary constants.}$$

1. The complete solution of the DE is given by $y = \text{C.F.} + \text{P.I.}$

To find the C.F. :

If the roots of the auxiliary equation are,

(i) real and distinct say $m_1 \neq m_2 \neq \dots \neq m_n$ C.F $y = c_1 e^{m_1 x} + c_2 e^{m_2 x} + \dots + c_n e^{m_n x}$

(ii) real and equal say $m_1 = m_2 = m$

$$\text{C.F} = (c_1 x + c_2) e^{mx} + c_3 e^{m_3 x} + c_4 e^{m_4 x} + \dots + c_n e^{m_n x}$$

(iii) real and equal say $m_1 = m_2 = m_3 = m$,

$$\text{C.F} = (c_1 x^2 + c_2 x + c_3) e^{mx} + c_4 e^{m_4 x} + \dots + c_n e^{m_n x}$$

(iv) one pair of roots is complex say $m_1 = \alpha + i\beta$, $m_2 = \alpha - i\beta$ and the other roots are real and unequal $\text{C.F} = e^{\alpha x} (c_1 \cos \beta x + c_2 \sin \beta x) + c_3 e^{m_3 x} + \dots + c_n e^{m_n x}$

(v) two pairs of roots are complex

$$\text{C.F} = e^{\alpha x} [(c_1 x + c_2) \cos \beta x + (c_3 x + c_4) \sin \beta x] + c_5 e^{m_5 x} + \dots + c_n e^{m_n x}$$

Example 1: Solve $\frac{d^2 y}{dx^2} + 2 \frac{dy}{dx} - 24y = 0$

Solution:

The operator form of the given equation can be written as $(D^2 + 2D - 24)y = 0$

The auxiliary equation is $m^2 + 2m - 24 = 0$

$$\text{(i.e.) } (m + 6)(m - 4) = 0$$

$$\therefore m_1 = -6, \quad m_2 = 4$$

These roots are real and distinct.

Hence the solutions is $y = Ae^{-6x} + Be^{4x}$

Example 2: Solve $(D^3 - 6D^2 + 11D - 6)y = 0$

Solution:

The auxiliary equation is $m^3 - 6m^2 + 11m - 6 = 0$

$$\begin{array}{r|rrrr} 1 & 1 & -6 & 11 & 6 \\ & 0 & 1 & -5 & 6 \\ \hline & 1 & -5 & 6 & 0 \end{array}$$

$\therefore$ The one of the root is $m_1 = 1$

The reduced equation is $m^2 - 5m + 6 = 0$

$$(m-2)(m-3) = 0$$

$$\therefore m_2 = 2, \quad m_3 = 3$$

$\therefore$ All these roots are real and different.

Hence the solution is $y = Ae^x + Be^{2x} + Ce^{3x}$

Example 3: Solve $(D^2 - 16)y = 0$ **given that** $y = 2, \frac{dy}{dx} = 0$ **when** $x = 0$.

Solution:

The auxiliary equation is $m^2 - 16 = 0$

$$m^2 = 16$$

$$m = \pm 4$$

$\therefore$ The roots are $m_1 = 4, \quad m_2 = -4$

$$\therefore y = Ae^{4x} + Be^{-4x}$$

$$\text{(i.e.) } y(x) = Ae^{4x} + Be^{-4x} \quad \dots(1)$$

$$\frac{dy(x)}{dx} = 4Ae^{4x} - 4Be^{-4x}$$

Given that $y(0) = 2$

$$(1) \Rightarrow A + B = 2 \quad \dots(2)$$

$$\text{Also } y'(0) = 0, 4A - 4B = 0 \Rightarrow A = B$$

$$(2) \Rightarrow 2A = 2 \Rightarrow A = 1 \text{ \& } B = 1$$

$$\therefore \text{ The solution is } y = e^{4x} + e^{-4x}$$

Example 4: Solve $\frac{d^2 y}{dx^2} - 10 \frac{dy}{dx} + 25y = 0$

Solution:

$$\text{The given equation can be written as } (D^2 - 10D + 25)y = 0$$

$$\text{The A.E. is } m^2 - 10m + 25 = 0$$

$$(m - 5)^2 = 0$$

$$m = 5 \text{ (twice)}$$

$$\text{The roots are } m_1 = m_2 = 5$$

$$\text{Hence the solution is } y = (Ax + B)e^{5x}$$

Example 5: Solve $(D^3 - 3D^2 + 3D - 1)y = 0$

Solution:

$$\text{The A.E. is } m^3 - 3m^2 + 3m - 1 = 0$$

$$\text{(i.e.) } (m - 1)^3 = 0$$

$$m = 1 \text{ (thrice)}$$

$$\text{Hence the solution is } y = e^x (A + Bx + Cx^2)$$

Example 6: Solve $\frac{d^3 y}{dx^3} + \frac{d^2 y}{dx^2} - \frac{dy}{dx} - y = 0$

Solution:

$$\text{The given equation can be written as } (D^3 + D^2 - D - 1)y = 0$$

$$\text{The auxiliary equation is } m^3 + m^2 - m - 1 = 0$$

$$\text{(i.e.) } m^2(m + 1) - 1(m + 1) = 0$$

$$(m + 1)(m^2 - 1) = 0$$

$$\text{(i.e.) } (m + 1)(m + 1)(m - 1) = 0$$

$$m_1 = m_2 = -1, m_3 = 1$$

Hence the solution is $y = e^{-x}(A + Bx) + Ce^x$

Example 7: Solve $\frac{d^2y}{dx^2} - 6\frac{dy}{dx} + 13y = 0$

Solution:

The given equation is $(D^2 - 6D + 13)y = 0$

The auxiliary equation is $m^2 - 6m + 13 = 0$

$$m = \frac{6 \pm \sqrt{36 - 4(1)(13)}}{2} \quad \left[\therefore m = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \right]$$

$$= \frac{6 \pm \sqrt{-16}}{2} = 3 \pm 2i$$

The roots are imaginary and occur in conjugate pairs.

Hence the solution is $y = e^{3x}(A \cos 2x + B \sin 2x)$

Example 8: Solve $D^3 - D^2 - D - 2 = 0$

Solution:

The auxiliary equation is $m^3 - m^2 - m - 2 = 0$

$$2 \left| \begin{array}{cccc|c} 1 & -1 & -1 & -2 & \\ 0 & 2 & 2 & 2 & \\ 1 & 1 & 1 & 0 & \end{array} \right.$$

$\therefore$ The one of the roots is $m_1 = 2$

The reduced equation is $m^2 + m + 1 = 0$

$$m = \frac{-1 \pm \sqrt{1 - 4}}{2} = -\frac{1}{2} \pm i \frac{\sqrt{3}}{2}$$

One root is real and the remaining two are conjugate imaginaries.

Hence the solution is $y = Ae^{2x} + e^{-\frac{1}{2}x} \left(B \cos \frac{\sqrt{3}}{2}x + C \sin \frac{\sqrt{3}}{2}x \right)$

Example 9: Solve $(D^4 + 4D^3 + 8D^2 + 8D + 4)y = 0$

Solution: The auxiliary equation is $m^4 + 4m^3 + 8m^2 + 8m + 4 = 0$

$$\text{(i.e.) } (m^2 + 2m + 2)^2 = 0$$

$$(m^2 + 2m + 2)((m^2 + 2m + 2) = 0$$

$$m = \frac{-2 \pm \sqrt{4-8}}{2} \text{ (twice)}$$

$$\text{(i.e.) } m = -1 \pm i \text{ twice}$$

Here the roots are imaginary as well as repeated

Hence the solution is $y = e^{-x} [(A + Bx) \cos x + (C + Dx) \sin x]$

Exercise

Solve the following equations:

1. $3 \frac{d^2 y}{dx^2} + 10 \frac{dy}{dx} - 8y = 0$

6. $\frac{d^3 y}{dx^3} + \frac{d^2 y}{dx^2} - 5 \frac{dy}{dx} + 3y = 0$

2. $5 \frac{d^2 x}{dt^2} + 16 \frac{dx}{dt} + 12x = 0$

7. $\frac{d^3 y}{dx^3} - 2 \frac{d^2 y}{dx^2} + 4 \frac{dy}{dx} - 8y = 0$

3. $\frac{d^2 y}{dx^2} - a^2 y = 0$

8. $(D^4 + 2D^2 + 1)y = 0$

4. $\frac{d^2 y}{dx^2} - 4 \frac{dy}{dx} + 4y = 0$

9. $(D^4 + 3D^2 - 4)y = 0$

10. $(D^3 + 9D^2 + 23D + 15)y = 0$

5. $\frac{d^2 y}{dx^2} - \frac{dy}{dx} + y = 0$

Answers

:

1. $y = Ae^{\frac{2}{3}x} + Be^{-4x}$

4. $y = e^{2x}(A + Bx)$

2. $y = Ae^{-2t} + Be^{\frac{-6}{5}t}$

5. $y = e^{\frac{x}{2}} \left[A \cos \frac{\sqrt{3}}{2} x + B \sin \frac{\sqrt{3}}{2} x \right]$

3. $y = Ae^{ax} + Be^{-ax}$

6. $y = e^{2x}(A + Bx) + Ce^{-3x}$

$$7. y = Ae^{2x} + B \cos 2x + C \sin 2x$$

$$9. y = Ae^x + Be^{-x} + C \cos 2x + D \sin 2x$$

$$8. y = (A + Bx) \cos x + (C + Dx) \sin x$$

$$10. y = Ae^{-x} + Be^{-5x} + Ce^{-3x}$$

Particular Integral

The particular integral (P.I) of the solution of the equation $f(D)y = X \dots (2)$ in the function v , where $y = v$ is a particular solution of (2) containing no arbitrary constants. The particular integral depends on the function X and is defined as

$$P.I = \frac{1}{f(D)} X, \text{ where } \frac{1}{f(D)} \text{ is the inverse operator of } f(D).$$

$$\text{(i.e.) } f(D) \left\{ \frac{1}{f(D)} X \right\} = X$$

Type I

If $X = e^{ax}$, where a is constant.

$$P.I = \frac{1}{f(D)} e^{ax} = \frac{1}{f(a)} e^{ax}, \text{ if } f(a) \neq 0$$

$$\text{If } f(a) = 0, \text{ then } P.I = x \cdot \frac{1}{f'(D)} e^{ax} = x \frac{1}{f'(a)} e^{ax} \text{ provided } f'(a) \neq 0$$

$$\text{If } f'(a) = 0, \text{ then } P.I = x^2 \frac{1}{f''(D)} e^{ax} = x^2 \frac{1}{f''(a)} e^{ax}, \text{ provided } f''(a) \neq 0$$

and this process has to be continued till a non zero value is obtained in the denominator.

Example 1: Find the P.I of $(D^2 - 5D + 6)y = e^{4x}$

Solution:

The given equation is $(D^2 - 5D + 6)y = e^{4x}$

$$\begin{aligned} P.I. &= \frac{1}{D^2 - 5D + 6} e^{4x} \\ &= \frac{1}{4^2 - 5(4) + 6} \cdot e^{4x} \quad (\text{Replacing D by 4}) \\ &= \frac{1}{2} e^{4x} \end{aligned}$$

Example 2: Solve $\frac{d^2y}{dx^2} + 4y = e^{-2x}$

Solution:

The given equation is $(D^2 + 4)y = e^{-2x}$

The auxiliary equation is $m^2 + 4 = 0$

$$m^2 = -4$$

$$m = \pm 2i$$

N

$$\alpha = 0, \quad \beta = 2$$

Complementary Function $= e^{0x} (A \cos 2x + B \sin 2x)$

$$(\text{i.e.}) \quad C.F. = A \cos 2x + B \sin 2x$$

$$\text{Particular integral} = \frac{1}{D^2 + 4} e^{-2x} = \frac{1}{(-2)^2 + 4} e^{-2x} \quad (\text{Replacing D by -2})$$

$$= \frac{1}{8} e^{-2x}$$

$\therefore$ The complete solution is $y = C.F. + P.I$

$$y = A \cos 2x + B \sin 2x + \frac{1}{8} e^{-2x}$$

Example 3: Solve $(D^2 - 5D + 6)y = 3e^{5x}$

Solution:

The given equation is $(D^2 - 5D + 6)y = 3e^{5x}$

The auxiliary equation is $m^2 - 5m + 6 = 0$

$$(m-3)(m-2) = 0$$

$$m = 3 \text{ or } m = 2$$

Complementary Function = $Ae^{2x} + Be^{3x}$

$$\begin{aligned} \text{Particular Integral} &= \frac{1}{D^2 - 5D + 6} 3e^{5x} = \frac{1}{5^2 - 5(5) + 6} 3e^{5x} && \text{(Replacing D by 5)} \\ &= \frac{1}{6} 3e^{5x} = \frac{1}{2} e^{5x} \end{aligned}$$

The complete solution is $y = C.F. + P.I.$

$$y = Ae^{2x} + Be^{3x} + \frac{e^{5x}}{2}$$

Example 4: Solve $(D^2 + 2D + 1)y = e^{-x} + 3$

Solution:

The given equation is $(D^2 + 2D + 1)y = e^{-x} + 3$

The auxiliary equation is $m^2 + 2m + 1 = 0$

$$(m+1)(m+1) = 0$$

$$m = -1, -1$$

Complementary Function = $e^{-x}(Ax + B)$

$$\text{Particular Integral} = \frac{1}{D^2 + 2D + 1} (e^{-x} + 3)$$

$$P.I._1 = \frac{1}{D^2 + 2D + 1} e^{-x}$$

$$= \frac{1}{(-1)^2 + 2(-1) + 1} e^{-x}$$

(Replacing D by -1)

$$= \frac{1}{1-2+1} e^{-x} = \frac{x}{2D+2} e^{-x} \quad (\because \text{Denominator is } 0)$$

$$= \frac{x}{2(-1)+2} e^{-x}$$

$$\therefore P.I_1 = \frac{x^2}{2} e^{-x} \quad (\because \text{Denominator is } 0)$$

$$P.I_2 = \frac{1}{D^2 + 2D + 1} \cdot 3 = \frac{1}{D^2 + 2D + 1} 3e^{0x}$$

$$= \frac{1}{(0)^2 + 2(0) + 1} 3e^{0x} = 3$$

(Replacing D by 0)

$\therefore$ The complete solution is $y = C.F. + P.I_1 + P.I_2$

$$(\text{i.e.}) \quad y = e^{-x}(Ax + B) + \frac{x^2}{2} e^{-x} + 3$$

Example 5: Solve $(D^2 - 6D + 9)y = e^{3x}$ **(Jan - 2010)**

Solution:

The given equation is $(D^2 - 6D + 9)y = e^{3x}$

The auxiliary equation is $m^2 - 6m + 9 = 0$

$$(m-3)^2 = 0$$

$$m = 3 \quad (\text{twice})$$

Complementary Function $= e^{3x}(Ax + B)$

$$\text{Particular Integral} = \frac{1}{(D-3)^2} e^{3x}$$

$$= \frac{x}{2(D-3)} e^{3x} \quad (\because \text{Denominator is } 0)$$

$$= \frac{x^2}{2} e^{3x} \quad (\because \text{Denominator is 0})$$

$\therefore$ The complete solution is $y = C.F + P.I$

$$(\text{i.e.}) \quad y = e^{3x}(Ax + B) + \frac{x^2 e^{3x}}{2}$$

Example 6: Solve $(D^3 + 3D^2 + 3D + 1)y = e^{-x}$

Solution:

The given differential equation is $(D^3 + 3D^2 + 3D + 1)y = e^{-x}$

The auxiliary equation is $m^3 + 3m^2 + 3m + 1 = 0$

$$(m+1)^3 = 0$$

$$(m+1)(m+1)(m+1) = 0$$

$$\therefore m = -1, -1, -1$$

$\therefore$ Complementary function $= (Ax^2 + Bx + C)e^{-x}$

$$\begin{aligned} \text{Particular integral} &= \frac{1}{D^3 + 3D^2 + 3D + 1} e^{-x} \\ &= \frac{1}{(-1)^3 + 3(-1)^2 + 3(-1) + 1} e^{-x} \end{aligned}$$

(Replacing D by -1)

$$= \frac{x}{3D^2 + 6D + 3} e^{-x} \quad (\because \text{Denominator is 0})$$

$$= \frac{x}{6-6} e^{-x}$$

$$= \frac{x^2}{6D+6} e^{-x} \quad (\because \text{Denominator is 0})$$

$$= \frac{x^2}{6-6} e^{-x}$$

(Replacing D by -1)

$$\therefore P.I_1 = \frac{x^3}{6} e^{-x}$$

$\therefore$ The complete solution is $y = C.F. + P.I.$

$$y = (Ax^2 + Bx + C) + \frac{x^3}{6} e^{-x}$$

Example 7: Solve $(D^2 - 7D + 12)y = \sinh 2x$

Solution:

The given differential equation is $(D^2 - 7D + 12)y = \sinh 2x$

The auxiliary equation is $m^2 - 7m + 12 = 0$

$$(m-3)(m-4) = 0$$

$$m-3=0 \text{ (or) } m-4=0$$

$$m_1 = 3, m_2 = 4$$

$\therefore$ Complementary function $= Ae^{3x} + Be^{4x}$

$$\text{Particular Integral} = \frac{1}{D^2 - 7D + 12} \sinh 2x$$

$$= \frac{1}{D^2 - 7D + 12} \left[\frac{e^{2x} - e^{-2x}}{2} \right]$$

$$= \frac{1}{2} \left[\frac{1}{D^2 - 7D + 12} e^{2x} - \frac{1}{D^2 - 7D + 12} e^{-2x} \right]$$

$$= \frac{1}{2} \left[\frac{1}{2^2 - 7(2) + 12} e^{2x} - \frac{1}{(-2)^2 - 7(-2) + 12} e^{-2x} \right]$$

$$= \frac{1}{2} \left[\frac{1}{4 - 14 + 12} e^{2x} - \frac{1}{4 + 14 + 12} e^{-2x} \right]$$

$$= \frac{1}{2} \left[\frac{e^{2x}}{2} - \frac{e^{-2x}}{30} \right]$$

$$\therefore P.I. = \frac{e^{2x}}{4} - \frac{e^{-2x}}{60}$$

$\therefore$ The complete solution is $y = C.F. + P.I.$

$$y = Ae^{3x} + Be^{4x} + \frac{e^{2x}}{4} - \frac{e^{-2x}}{60}$$

Example 8: Solve $\frac{d^2y}{dx^2} + 5\frac{dy}{dx} + 6y = 3e^{-2x} + e^{-3x}$ **(Apr - 2003)**

Solution:

The given differential equation is $(D^2 + 5D + 6)y = 3e^{-2x} + e^{-3x}$

The auxiliary equation is $m^2 + 5m + 6 = 0$

$$(m+3)(m+2) = 0$$

$$m_1 = -3, m_2 = -2$$

$\therefore$ Complementary function $= Ae^{-3x} + Be^{-2x}$

$$\text{Particular Integral} = \frac{1}{D^2 + 5D + 6} (3e^{-2x} + e^{-3x})$$

$$P.I_1 = \frac{1}{D^2 + 5D + 6} 3e^{-2x} \quad (\text{Replacing } D \text{ by } -2)$$

$$= \frac{1}{(-2)^2 + 5(-2) + 6} 3e^{-2x}$$

$$= \frac{1}{10 - 10} 3e^{-2x}$$

$$= \frac{3x}{2D + 5} e^{-2x} \quad (\because \text{Denominator is } 0)$$

$$= \frac{3x}{2(-2) + 5} e^{-2x}$$

$$\therefore P.I_1 = 3e^{-2x}$$

$$P.I_2 = \frac{1}{D^2 + 5D + 6} e^{-3x}$$

$$= \frac{1}{9 - 15 + 6} e^{-3x} \quad (\text{Replacing } D \text{ by } -3)$$

$$= \frac{x}{2D + 5} e^{-3x} \quad (\because \text{Denominator is } 0)$$

$$P.I_2 = -xe^{-3x} \quad (\text{Replacing } D \text{ by } -3)$$

$\therefore$ The complete solution is $y = C.F + P.I_1 + P.I_2$

$$y = Ae^{-3x} + Be^{-2x} + 3xe^{-2x} - xe^{-3x}$$

Exercise

Solve the following equations

1. $(D^2 - 3D - 4)y = e^{2x}$

7. $(D^5 - D)y = 12e^x$

2. $(D^2 - 7D + 12)y = e^x$

8. $(D^2 + 2D + 2)y = \sinh x$

3. $(D^2 + 2D + 2)y = 3 + 2e^x$

9. $(D^2 - 2D + 1)y = \cosh x + 5e^{3x}$

4. $(D^2 - 4D + 4)y = e^{2x}$

10. $(D^2 + 2D + 1)y = 2e^{3x}$

5. $\frac{d^2 y}{dx^2} + 6\frac{dy}{dx} + 9y = 5e^{3x}$

6. $(D^3 - 3D^2 + 4D - 2)y = e^x$

Answers

1. $y = Ae^{2x} + Be^{-x} - \frac{1}{4}e^{2x}$

2. $y = Ae^{4x} + Be^{3x} + \frac{1}{4}e^x$

3. $y = e^{-x}(A \cos x + B \sin x) + \frac{3}{2} + \frac{2e^x}{5}$

$$4. y = (A + Bx)e^{2x} + \frac{x^2}{2}e^{2x}$$

$$5. y = Ae^{-x} + Be^{-5x} + \frac{1}{21}e^{2x}$$

$$6. y = Ae^x + e^x [B \cos x + C \sin x] + xe^x$$

$$7. y = C_1 + C_2 e^x + C_3 e^{-x} + [C_4 \cos x + C_5 \sin x] + 3xe^x$$

$$8. y = e^{-x} [A \cos x + B \sin x] + \frac{e^{-x}}{10} - \frac{e^{-x}}{2}$$

$$9. y = (A + Bx)e^x + \frac{x^2}{4}e^x + \frac{x^{-x}}{8} + \frac{5e^{3x}}{4}$$

$$10. y = e^{-x}(Ax + B) + \frac{e^{3x}}{8}$$

Type II

If $X = \sin ax$ (or) $\cos ax$, where a is constant.

$$P.I. = \frac{1}{f(D)} \sin ax \text{ (or) } \cos ax$$

In $f(D)$, replace D^2 by $-a^2$, provided $f(D) \neq 0$ when we replace D^2 by $-a^2$. If $f(D) = 0$, then

$$P.I. = x \frac{1}{f'(D)} \sin ax \text{ (or) } \cos ax$$

Again replace D^2 by $-a^2$ in $f'(D)$ provided $f'(D) \neq 0$. If $f'(D) = 0$, then

$$P.I. = x^2 \frac{1}{f''(D)} \sin ax \text{ (or) } \cos ax$$

Again replace D^2 by $-a^2$, provided $f''(D) \neq 0$, and this process should be continued till a non zero value is obtained in the denominator.

Example 1: Solve $\frac{d^2y}{dx^2} + 3\frac{dy}{dx} + 2y = \sin 3x$

Solution:

The given differential equation is $(D^2 + 3D + 2)y = \sin 3x$

The auxiliary equation is $m^2 + 3m + 2 = 0$

$$(m+1)(m+2) = 0$$

$$m_1 = -1, \quad m_2 = -2$$

Complementary function $= Ae^{-x} + Be^{-2x}$

$$\text{Particular Integral} = \frac{1}{D^2 + 3D + 2} \cdot \sin 3x$$

$$= \frac{1}{-(3)^2 + 3D + 2} \sin 3x \quad (\text{Replacing } D^2 \text{ by } -3^2)$$

$$= \frac{1}{3D - 7} \sin 3x$$

$$= \frac{3D + 7}{(3D - 7)(3D + 7)} \sin 3x$$

$$= \frac{3D + 7}{9D^2 - 49} \sin 3x$$

$$= \frac{3D + 7}{-81 - 49} \sin 3x \quad (\text{Replacing } D^2 \text{ by } -3^2)$$

$$= \frac{3D(\sin 3x) + 7(\sin 3x)}{-130}$$

$$P.I. = \frac{9 \cos 3x + 7 \sin 3x}{-130}$$

$\therefore$ The complete solution is $y = C.F + P.I$

$$y = Ae^{-x} + Be^{-2x} - \frac{1}{130}(9\cos 3x + 7\sin 3x)$$

Example 2: Solve $(D^2 + 2D + 1)y = \cos 2x$

Solution:

The given differential equation is $(D^2 + 2D + 1)y = \cos 2x$

The auxiliary equation is $m^2 + 2m + 1 = 0$

$$(m+1)^2 = 0$$

$$m = -1 \text{ (twice)}$$

$\therefore$ Complementary function $= (Ax + B)e^{-x}$

$$\text{Particular Integral} = \frac{1}{D^2 + 2D + 1} \cos 2x$$

$$= \frac{1}{-4 + 2D + 1} \cos 2x \quad (\text{Replacing } D^2 \text{ by } -2^2)$$

$$= \frac{1}{2D - 3} \cos 2x = \frac{1}{2D - 3} \times \frac{2D + 3}{2D + 3} \cos 2x$$

$$= \frac{(2D + 3) \cos 2x}{4D^2 - 9} = \frac{2D(\cos 2x) + 3 \cos 2x}{4(-4) - 9} \quad (\text{Replacing } D^2 \text{ by } -2^2)$$

$$= \frac{4(-\sin 2x) + 3 \cos 2x}{-16 - 9} = \frac{3 \cos x - 4 \sin 2x}{-25}$$

$$\therefore P.I. = \frac{4 \sin 2x - 3 \cos 2x}{25}$$

The complete solution is $y = C.F + P.I$

$$y = (Ax + B)e^{-x} + \frac{4 \sin 2x - 3 \cos 2x}{25}$$

Example 3: Solve $(D^2 - 4D + 3)y = \sin 3x \cos 2x$

Solution:

The given differential equation is $(D^2 - 4D + 3)y = \sin 3x \cos 2x$

The auxiliary equation is $m^2 - 4m + 3 = 0$

$$(m-1)(m-3) = 0$$

$$m_1 = 1, \quad m_2 = 3$$

$\therefore$ The complementary function $= Ae^x + Be^{3x}$

$$\text{Particular Integral} = \frac{1}{(D^2 - 4D + 3)} \sin 3x \cos 2x$$

$$= \frac{1}{2} \cdot \frac{1}{D^2 - 4D + 3} (\sin 5x + \sin x)$$

$$\left(\because \sin 3x \cos 2x = \frac{1}{2} (\sin 5x + \sin x) \right)$$

$$= \frac{1}{2} \left[\frac{1}{D^2 - 4D + 3} \sin 5x + \frac{1}{D^2 - 4D + 3} \sin x \right]$$

$$P.I._1 = \frac{1}{2} \frac{1}{D^2 - 4D + 3} \sin 5x$$

$$= \frac{1}{2} \frac{1}{(-25) - 4D + 3} \sin 5x \quad (\text{Replacing } D^2 \text{ by } -5^2)$$

$$= \frac{1}{2} \frac{1}{-(4D + 22)} \sin 5x$$

$$= -\frac{1}{4} \cdot \frac{1}{(2D + 11)} \sin 5x$$

$$= -\frac{1}{4} \frac{2D - 11}{(2D - 11)(2D + 11)} \cdot \sin 5x$$

$$= -\frac{1}{4} \frac{(2D - 11)}{(4D^2 - 121)} \sin 5x$$

$$= -\frac{1}{4} \frac{2D(\sin 5x) - 11(\sin 5x)}{4(-25) - 121} \quad (\text{Replacing } D^2 \text{ by } -5^2)$$

$$\begin{aligned}
&= -\frac{1}{4} \cdot \frac{2(5\cos 5x) - 11\sin 5x}{-221} \\
\therefore P.I_1 &= \frac{10\cos 5x - 11\sin 5x}{884} \\
P.I_2 &= \frac{1}{2} \cdot \frac{1}{D^2 - 4D + 3} \sin x \\
&= \frac{1}{2} \cdot \frac{1}{(-1) - 4D + 3} \sin x \quad (\text{Replacing } D^2 \text{ by } -1^2) \\
&= \frac{1}{2} \cdot \frac{1}{2 - 4D} \sin x = \frac{1}{4} \cdot \frac{1}{1 - 2D} \sin x \\
&= \frac{1}{4} \cdot \frac{1 + 2D}{(1 - 2D)(1 + 2D)} \cdot \sin x \\
&= \frac{1}{4} \cdot \frac{1 + 2D}{(1 - 4D^2)} \cdot \sin x \\
&= \frac{1}{4} \cdot \frac{1 + 2D}{1 - 4(-1)} \cdot \sin x \quad (\text{Replacing } D^2 \text{ by } -1^2) \\
\therefore P.I_2 &= \frac{\sin x + 2\cos x}{20}
\end{aligned}$$

The complete solution is $y = C.F + P.I_1 + P.I_2$

$$y = Ae^x + Be^{3x} + \frac{10\cos 5x - 11\sin 5x}{884} + \frac{\sin x + 2\cos x}{20}$$

Example 4: Solve $(D^2 + 5D + 6)y = \cos^2 x$ **(May- 2011)**

Solution:

The given differential equation is $(D^2 + 5D + 6)y = \cos^2 x$

The auxiliary equation is $m^2 + 5m + 6 = 0$

$$(m + 2)(m + 3) = 0$$

$$m_1 = -2, \quad m_2 = -3$$

$$\therefore C.F = Ae^{-2x} + Be^{-3x}$$

$$\begin{aligned} \text{Particular Integral} &= \frac{1}{D^2 + 5D + 6} \cdot \left(\frac{1 + \cos 2x}{2} \right) \quad \left(\because \cos^2 x = \frac{1 + \cos 2x}{2} \right) \\ &= \frac{1}{D^2 + 5D + 6} \cdot \frac{e^{0x}}{2} + \frac{1}{D^2 + 5D + 6} \cdot \frac{\cos 2x}{2} \\ &= \frac{1}{12} + \frac{1}{-(2)^2 + 5D + 6} \cdot \frac{\cos 2x}{2} \quad (\text{Replacing } D^2 \text{ by } -2^2) \\ &= \frac{1}{12} + \frac{1}{5D + 2} \cdot \frac{\cos 2x}{2} \\ &= \frac{1}{12} + \frac{1}{2} \cdot \frac{5D - 2}{(5D + 2)(5D - 2)} \cdot \cos 2x = \frac{1}{12} + \frac{1}{2} \cdot \frac{5D(\cos 2x) - 2 \cos 2x}{25D^2 - 4} \\ &= \frac{1}{12} + \frac{1}{2} \cdot \left(\frac{5(-2 \sin 2x) - 2 \cos 2x}{25(-4) - 4} \right) = \frac{1}{12} + \frac{1}{2} \cdot \left(\frac{-10 \sin 2x - 2 \cos 2x}{-104} \right) \\ \therefore P.I. &= \frac{1}{12} + \frac{5 \sin 2x + \cos 2x}{104} \end{aligned}$$

$\therefore$ The complete solution is $y = C.F. + P.I.$

$$y = Ae^{-2x} + Be^{-3x} + \frac{1}{12} + \frac{5 \sin 2x + \cos 2x}{104}$$

Example 5: Solve $(D^2 + 3D + 2)y = e^{2x} + \sin x$

(Nov- 2011)

Solution:

The given differential equation is $(D^2 + 3D + 2)y = e^{2x} + \sin x$

The auxiliary equation is $m^2 + 3m + 2 = 0$

$$(m+1)(m+2) = 0$$

$$m_1 = -1, \quad m_2 = -2$$

$$\therefore C.F. = Ae^{-x} + Be^{-2x}$$

$$\text{Particular Integral} = \frac{1}{D^2 + 3D + 2} (e^{2x} + \sin x)$$

$$\begin{aligned} P.I_1 &= \frac{1}{D^2 + 3D + 2} e^{2x} \\ &= \frac{1}{4 + 6 + 2} e^{2x} \quad (\text{Replacing } D \text{ by } 2) \end{aligned}$$

$$P.I_1 = \frac{e^{2x}}{12}$$

$$\begin{aligned} P.I_2 &= \frac{1}{D^2 + 3D + 2} \sin x \\ &= \frac{1}{-1 + 3D + 2} \sin x \quad (\text{Replacing } D^2 \text{ by } -1^2) \end{aligned}$$

$$\begin{aligned} &= \frac{1}{3D + 1} \sin x = \frac{3D - 1}{(3D + 1)(3D - 1)} \sin x \\ &= \frac{3D(\sin x) - \sin x}{9D^2 - 1} \quad (\text{Replacing } D^2 \text{ by } -1^2) \end{aligned}$$

$$= \frac{3\cos x - \sin x}{-9 - 1} \quad (\text{Replacing } D^2 \text{ by } -1^2)$$

$$P.I_2 = \frac{3\cos x - \sin x}{-10}$$

The complete solution is $y = C.F. + P.I_1 + P.I_2$.

$$y = Ae^{-x} + Be^{-2x} + \frac{e^{2x}}{12} + \frac{3\cos x - \sin x}{-10}$$

Example 6: Solve $(D^2 + 6D + 8)y = e^{-2x} + \cos^2 x$

Solution:

The given differential equation is $(D^2 + 6D + 8)y = e^{-2x} + \cos^2 x$

The auxiliary equation is $m^2 + 6m + 8 = 0$

$$(m+4)(m+2)=0$$

$$m_1 = -4, \quad m_2 = -2$$

$$\text{Complementary Function} = Ae^{-4x} + Be^{-2x}$$

$$\text{Particular Integral} = \frac{1}{D^2 + 6D + 8} (e^{-2x} + \cos^2 x)$$

$$= \frac{1}{D^2 + 6D + 8} e^{-2x} + \frac{1}{D^2 + 6D + 8} \cos^2 x$$

$$P.I_1 = \frac{1}{D^2 + 6D + 8} e^{-2x}$$

$$= \frac{1}{4 - 12 + 8} e^{-2x} \quad (\text{Replacing } D \text{ by } -2)$$

$$= \frac{x}{2D + 6} e^{-2x} \quad (\because D \text{ is } 0)$$

$$= \frac{x}{-4 + 6} e^{-2x} = \frac{x}{2} e^{-2x}$$

$$P.I_2 = \frac{1}{D^2 + 6D + 8} \left[\frac{1 + \cos 2x}{2} \right]$$

$$= \frac{1}{D^2 + 6D + 8} \cdot \frac{e^{0x}}{2} + \frac{1}{D^2 + 6D + 8} \frac{\cos 2x}{2}$$

$$= \frac{1}{16} + \frac{1}{2} \left[\frac{\cos 2x}{-4 + 6D + 8} \right] \quad (\text{Replacing } D^2 \text{ by } -2^2)$$

$$= \frac{1}{16} + \frac{1}{2} \left[\frac{\cos 2x}{6D + 4} \right] = \frac{1}{16} + \frac{1}{2} \left[\frac{(6D - 4) \cos 2x}{(6D + 4)(6D - 4)} \right]$$

$$= \frac{1}{16} + \frac{1}{2} \left[\frac{6D(\cos 2x) - 4 \cos 2x}{36D^2 - 16} \right] = \frac{1}{16} + \frac{1}{2} \left[\frac{-12 \sin 2x - 4 \cos 2x}{36(-4) - 16} \right]$$

$$= \frac{1}{16} + \frac{1}{2} \left[\frac{-12 \sin 2x - 4 \cos 2x}{-160} \right] = \frac{1}{16} + \left(\frac{6 \sin 2x + 2 \cos 2x}{160} \right)$$

$$\therefore P.I_2 = \frac{1}{16} + \frac{3\sin 2x + \cos 2x}{80}$$

$\therefore$ The complete solution is $y = C.F. + P.I_1 + P.I_2$.

$$y = Ae^{-4x} + Be^{-2x} + \frac{x}{2}e^{-2x} + \frac{1}{16} + \frac{3\sin 2x + \cos 2x}{80}$$

Example 7: Solve $(D^2 + 16)y = e^{-3x} + \cos 4x$

Solution:

The given differential equation is $(D^2 + 16)y = e^{-3x} + \cos 4x$

The auxiliary equation is $m^2 = -16$

$$m = \pm 4i$$

$$\therefore C.F. = A\cos 4x + B\sin 4x$$

$$\text{Particular Integral } P.I. = \frac{1}{D^2 + 16}(e^{-3x} + \cos 4x)$$

$$P.I_1 = \frac{1}{D^2 + 16}e^{-3x}$$

$$= \frac{1}{9 + 16}e^{-3x} \quad (\text{Replacing } D \text{ by } -3)$$

$$P.I_1 = \frac{e^{-3x}}{25}$$

$$P.I_2 = \frac{1}{D^2 + 16} \cdot \cos 4x$$

$$= \frac{1}{-16 + 16} \cdot \cos 4x \quad (\text{Replacing } D^2 \text{ by } -4^2)$$

$$= \frac{x}{2D} \cdot \cos 4x = \frac{x}{2} \int \cos 4x dx$$

$$= \frac{x}{2} \frac{\sin 4x}{4} = \frac{x \sin 4x}{8}$$

$\therefore$ The complete solution is $y = C.F. + P.I_1 + P.I_2$.

$$y = A \cos 4x + B \sin 4x + \frac{e^{-3x}}{25} + \frac{x \sin 4x}{8}$$

Example 8: Solve $(D^2 + 2D - 3)y = e^{2x} + \sin 2x$

Solution:

The auxiliary equation is $m^2 + 2m - 3 = 0$

$$(m-1)(m+3) = 0$$

$$m_1 = -3, \quad m_2 = 1$$

$$\therefore C.F = Ae^{-3x} + Be^x$$

$$\text{Particular Integral} = \frac{1}{D^2 + 2D - 3} (e^{2x} + \sin 2x)$$

$$P.I_1 = \frac{1}{D^2 + 2D - 3} \cdot e^{2x}$$

$$= \frac{1}{4 + 4 - 3} \cdot e^{2x} \quad (\text{Replacing } D \text{ by } 2)$$

$$P.I_1 = \frac{e^{2x}}{5}$$

$$P.I_2 = \frac{1}{D^2 + 2D - 3} \sin 2x$$

$$= \frac{1}{-4 + 2D - 3} \sin 2x \quad (\text{Replacing } D^2 \text{ by } -4^2)$$

$$= \frac{(2D + 7) \sin 2x}{(2D - 7)(2D + 7)} = \frac{2D(\sin 2x) + 7 \sin 2x}{4D^2 - 49}$$

$$= \frac{4 \cos 2x + 7 \sin 2x}{4(-4) - 49} \quad (\text{Replacing } D^2 \text{ by } -2^2)$$

$$P.I_2 = \frac{4 \cos 2x + 7 \sin 2x}{-65}$$

$\therefore$ The complete solution is $y = C.F + P.I$

$$y = Ae^{-3x} + Be^x + \frac{e^{2x}}{5} + \frac{4 \cos 2x + 7 \sin 2x}{-65}$$

Exercise

Solve the following differential equations

1. $(D^2 + D + 1)y = \sin 2x$

6. $(D^2 + 5D - 6)y = \sin 4x \sin x$

2. $(D^2 - 3D + 2)y = \sin 5x$

7. $(D^2 - 4)y = \sin^2 x$

3. $(D^2 + 9)y = \cos 2x + 3$

8. $(3D^2 + D - 14)y = 8e^{2x} + \cos 5x$

4. $(D^2 + D + 2)y = e^x + \cos x$

9. $(D^2 - 3D + 2)y = \cos 3x \cos 2x$

5. $(D^2 - 4D - 5)y = e^x + 4 \cos 4x$

10. $(D^2 + 4D + 4)y = 4 \cos x + 3 \sin x$

Answers

1. $y = e^{-\frac{1}{2}x} \left(A \cos \frac{\sqrt{3}}{2} x + B \sin \frac{\sqrt{3}}{2} x \right) - \frac{1}{13} (2 \cos 2x + 3 \sin 2x)$

2. $y = Ae^x + Be^{2x} + \frac{1}{130} (9 \cos 3x - 7 \sin 3x)$

3. $y = A \cos 3x + B \sin 3x + \frac{1}{5} \cos 2x + \frac{1}{3}$

4. $y = e^{-\frac{x}{2}} \left(A \cos \frac{\sqrt{7}}{2} x + B \sin \frac{\sqrt{7}}{2} x \right) + \frac{e^x}{4} + \frac{\sin x + \cos x}{2}$

5. $y = Ae^{5x} + Be^{-x} - \frac{e^{3x}}{8} - \frac{1}{85} (14 \cos 3x + 12 \sin 3x)$

6. $y = Ae^{-5x} + Be^x \frac{\sin 3x - \cos 3x}{60} + \frac{31 \cos 5x - 25 \sin 5x}{3172}$

7. $y = Ae^{-2x} + Be^{2x} - \frac{1}{8} + \frac{1}{16} \cos 2x$

8. $y = Ae^{2x} + Be^{-\frac{7}{3}x} + \frac{8}{13} xe^{2x} + \frac{1}{7946} (5 \sin 5x - 89 \cos 5x)$

9. $y = Ae^x + Be^{2x} - \frac{(15 \sin x + 23 \cos 5x)}{1508} + \frac{\cos x + 3 \sin x}{20}$

10. $y = (Ax + B)e^{-2x} + \sin x$

Type III

When $x = x^m$, m being a positive integer (or) when $f(x)$ is an algebraic expression of the form $a_0x^n + a_1x^{n-1} + a_2x^{n-2} + \dots + a_n$, then,

$$P.I = \frac{1}{f(D)} x^m$$

Take out the lowest degree term from $f(D)$ to make the first term unity (so that Binomial theorem for a negative index is applicable). The remaining factor will be of the form $1 + \phi(D)$ or $1 - \phi(D)$. Take this factor in to the numerator. It takes the form $[1 + \phi(D)]^{-1}$ or $[1 - \phi(D)]^{-1}$. Expand it in

Binom $(1+x)$ $(1-x)$ $(1+x)$

ascending powers of D as far as the term containing D^m , since $D^{m+1}(x^m) = 0$, $D^{m+2}(x^m) = 0$ and so on... Operate on x^m term by term.

$$(1-x)^{-2} = 1 + 2x + 3x^2 + 4x^3 + \dots + \infty$$

Example1: Solve $(D^2 + D - 6)y = x$

Solution:

The given differential equation is $(D^2 + D - 6)y = x$

The auxiliary equation is $m^2 + m - 6 = 0$

$$(m+3)(m-2) = 0$$

$$m_1 = -3, \quad m_2 = 2$$

The complementary function $= Ae^{-3x} + Be^{2x}$

$$\text{Particular Integral} = \frac{1}{D^2 + D - 6} x$$

$$\begin{aligned}
&= \frac{1}{-6 \left[1 - \left(\frac{D^2 + D}{6} \right) \right]} x \\
&= -\frac{1}{6} \left[1 - \left(\frac{D^2 + D}{6} \right) \right]^{-1} x \\
&= -\frac{1}{6} \left(1 + \frac{D^2 + D}{6} \right) x \quad (\text{Neglect higher powers, } \because D^2(x) = 0) \\
&= -\frac{1}{6} \left(x + 0 + \frac{1}{6} \right) = -\frac{1}{36} (6x + 1)
\end{aligned}$$

$\therefore$ The complete solution is $y = C.F. + P.I.$

$$y = Ae^{-3x} + Be^{2x} - \frac{1}{36} (6x + 1)$$

Example 2: Solve $(D-1)^2 y = x$

(Jan- 2010)

Solution:

The given differential equation is $(D^2 - 2D + 1)y = x$

The auxiliary equation is $m^2 - 2m + 1 = 0$

$$(m-1)(m-1) = 0$$

$$m = 1(\text{twice})$$

$$\therefore C.F. = e^x (Ax + B)$$

$$\text{Particular Integral} = \frac{1}{D^2 - 2D + 1} x$$

$$= \frac{1}{1 + (D^2 - 2D)} x$$

$$= [1 + (D^2 - 2D)]^{-1} x$$

$$= [1 - (D^2 - 2D) + (D^2 - 2D)^2 \dots] x$$

$$= x + 2(1) \quad [\text{Neglecting higher powers of } D \text{ as } D^2(x) = 0]$$

∴ The complete solution is $y = C.F. + P.I.$

$$y = e^x (Ax + B) + x + 2$$

Example 3: Solve $(D^4 - 2D^3 + D^2)y = x^2 + e^x$

(May- 2012)

Solution:

The given differential equation is $(D^4 - 2D^3 + D^2)y = x^2 + e^x$

The auxiliary equation is $m^4 - 2m^3 + m^2 = 0$

$$m^2(m^2 - 2m + 1) = 0$$

$$m^2 = 0 \quad (\text{or}) \quad m^2 - 2m + 1 = 0$$

$$m = 0, \quad (\text{or}) \quad (m-1)(m-1) = 0$$

$$m = 1 \quad (\text{twice})$$

$$\therefore C.F. = (Ax + B) + (Cx + D)e^x$$

$$\text{Particular Integral} = \frac{1}{D^4 - 2D^3 + D^2} (x^2 + e^x)$$

$$P.I._1 = \frac{1}{D^2(D^2 - 2D + 1)} x^2$$

$$= \frac{1}{D^2(D-1)^2} x^2$$

$$= \frac{1}{D^2} [1 - D]^{-2} x^2$$

$$= \frac{1}{D^2} (1 + 2D + 3D^2 + 4D^3 + 5D^4) x^2$$

$$= \left(\frac{1}{D^2} + \frac{2}{D} + 3 + 4D + 5D^2 \right) x^2$$

$$= \frac{x^4}{12} + \frac{2x^3}{3} + 3x^2 + 8x + 10$$

$$\begin{aligned}
 P.I_2 &= \frac{1}{D^4 - 2D^3 + D^2} e^x \\
 &= \frac{1}{1 - 2 + 1} e^x \quad (\text{Replacing } D \text{ by } 1)
 \end{aligned}$$

Since the denominator is 0, multiply by x and differentiate the denominator with respect to D , we get,

$$\begin{aligned}
 P.I_2 &= \frac{xe^x}{4D^3 - 6D^2 + 2D} \\
 &= \frac{xe^x}{4 - 6 + 2} \\
 &= \frac{x^2 e^x}{12D^2 - 12D + 2} \\
 P.I_2 &= \frac{x^2 e^x}{2} \quad (\text{Replacing } D \text{ by } 1)
 \end{aligned}$$

$\therefore$ The complete solution is $y = C.F. + P.I.$

$$y = (Ax + B) + (Cx + D)e^x + \frac{x^4}{12} + \frac{2x^3}{3} + 3x^2 + 8x + 10 + \frac{x^2}{2} e^x$$

Example 4: Solve the equation $(D^2 - 4D + 3)y = \sin 3x + x^2$

Solution:

The given differential equation is $(D^2 - 4D + 3)y = \sin 3x + x^2$

The auxiliary equation is $m^2 - 4m + 3 = 0$

$$(m - 1)(m - 3) = 0$$

$$m_1 = 1, \quad m_2 = 3$$

$$\therefore C.F. = Ae^x + Be^{3x}$$

$$\text{Particular Integral} = \frac{1}{D^2 - 4D + 3} (\sin 3x + x^2)$$

$$\begin{aligned}
P.I_1 &= \frac{1}{D^2 - 4D + 3} \sin 3x \\
&= \frac{1}{-9 - 4D + 3} \sin 3x \quad (\text{Replacing } D^2 \text{ by } -3^2) \\
&= \frac{1}{-(4D + 6)} \sin 3x = -\frac{4D - 6}{(4D + 6)(4D - 6)} \sin 3x \\
&= -\frac{4(D \sin 3x) - 6 \sin 3x}{16D^2 - 36} \\
&= -\frac{(12 \cos 3x) - 6 \sin 3x}{16(-9) - 36} \quad (\text{Replacing } D^2 \text{ by } -3^2) \\
&= -\frac{12 \cos 3x - 6 \sin 3x}{-180} \\
P.I_1 &= \frac{2 \cos 3x - \sin 3x}{30}
\end{aligned}$$

$$\begin{aligned}
P.I_2 &= \frac{1}{D^2 - 4D + 3} x^2 \\
&= \frac{1}{3 \left[1 - \left(\frac{4D - D^2}{3} \right) \right]} x^2 \\
&= \frac{1}{3} \left[1 - \left(\frac{4D - D^2}{3} \right) \right]^{-1} x^2 \\
&= \frac{1}{3} \left[1 + \frac{4D - D^2}{3} + \left(\frac{4D - D^2}{3} \right)^2 + \dots \right] x^2 \\
&= \frac{1}{3} \left[1 + \frac{4}{3}D - \frac{D^2}{3} + \frac{16D^2}{9} \right] x^2
\end{aligned}$$

(Neglecting the higher power as $D^3(x^2) = 0$)

$$= \frac{1}{3} \left[x^2 + \frac{8}{3}x - \frac{2}{3} + \frac{32}{9} \right]$$

$$P.I_2 = \frac{1}{3} \left[x^2 + \frac{8}{3}x + \frac{26}{9} \right]$$

$\therefore$ The complete solution is $y = C.F. + P.I.$

$$y = Ae^x + Be^{3x} + \frac{2\cos 3x - \sin 3x}{30} + \frac{1}{3} \left(x^2 + \frac{8}{3}x + \frac{26}{9} \right)$$

Example 5: Solve $\frac{d^2y}{dx^2} - 5\frac{dy}{dx} + 6y = x^2 + 3$

Solution:

The given differential equation $(D^2 - 5D + 6)y = x^2 + 3$

The auxiliary equation is $m^2 - 5m + 6 = 0$

$$(m-2)(m-3) = 0$$

$$m_1 = 2, \quad m_2 = 3$$

$$\therefore C.F = Ae^x + Be^{3x}$$

$$\text{Particular Integral} = \frac{1}{D^2 - 5D + 6} (x^2 + 3)$$

$$= \frac{1}{6 \left[1 - \left(\frac{5D - D^2}{6} \right) \right]} (x^2 + 3)$$

$$= \frac{1}{6} \left[1 - \left(\frac{5D - D^2}{6} \right) \right]^{-1} (x^2 + 3)$$

$$= \frac{1}{6} \left(1 + \frac{5D - D^2}{6} + \left(\frac{5D - D^2}{6} \right)^2 + \dots \right) (x^2 + 3)$$

$$\begin{aligned}
&= \frac{1}{6} \left(1 + \frac{5D}{6} - \frac{D^2}{6} + \frac{25D^2}{36} \right) (x^2 + 3) \\
&= \frac{1}{6} \left(x^2 + 3 + \frac{10x}{6} - \frac{2}{6} + \frac{50}{36} \right) = \frac{1}{6} \left(x^2 + \frac{10x}{6} + 3 - \frac{1}{3} + \frac{25}{18} \right) \\
&= \frac{1}{6} \left(\frac{18x^2 + 30x + 73}{18} \right) = \frac{1}{108} (18x^2 + 30x + 73)
\end{aligned}$$

$\therefore$ The complete solution is $y = C.F + P.I$

$$y = Ae^{2x} + Be^{3x} + \frac{1}{108} [18x^2 + 30x + 73]$$

Example 6: Solve $(D^2 + 5D + 4)y = x^2 + 7x + 9$

Solution:

The given differential equation $(D^2 + 5D + 4)y = x^2 + 7x + 9$

The auxiliary equation is $m^2 + 5m + 4 = 0$

$$(m+1)(m+4) = 0$$

$$m_1 = -1, \quad m_2 = -4$$

$$\therefore C.F = Ae^{-x} + Be^{-4x}$$

$$\text{Particular Integral} = \frac{1}{D^2 + 5D + 4} (x^2 + 7x + 9)$$

$$= \frac{1}{4 \left[1 + \left(\frac{5D + D^2}{4} \right) \right]} (x^2 + 7x + 9)$$

$$= \frac{1}{4} \left[1 + \left(\frac{5D + D^2}{4} \right) \right]^{-1} (x^2 + 7x + 9)$$

$$= \frac{1}{4} \left[1 - \left(\frac{5D + D^2}{4} \right) + \left(\frac{5D + D^2}{4} \right)^2 - \dots \right] (x^2 + 7x + 9)$$

$$\begin{aligned}
&= \frac{1}{4} \left[1 - \left(\frac{5D}{4} \right) - \left(\frac{D^2}{4} \right) + \left(\frac{25D^2}{16} \right) \right] (x^2 + 7x + 9) \\
&\quad \left[\because D^3(x^2 + 7x + 9) = 0 \right] \\
&= \frac{1}{4} \left[x^2 + 7x + 9 - \frac{5}{4}(2x + 7) - \frac{2}{4} + \frac{25}{16}(2) \right] \\
&= \frac{1}{4} \left[x^2 + 7x + 9 - \frac{5x}{2} - \frac{35}{4} - \frac{1}{2} + \frac{25}{8} \right] \\
&= \frac{1}{4} \left[x^2 + \frac{18}{4}x + \frac{23}{8} \right]
\end{aligned}$$

$\therefore$ The complete solution is $y = C.F + P.I$

$$y = Ae^{-x} + Be^{-4x} + \frac{1}{32}(8x^2 + 36x + 23)$$

Example 7: Solve $(D^2 + 2D + 2)y = \cos x + x^2 + 7$

Solution:

The given differential equation is $(D^2 + 2D + 2)y = \cos x + x^2 + 7$

The auxiliary equation is $m^2 + 2m + 2 = 0$

$$\begin{aligned}
m &= \frac{-2 \pm \sqrt{4 - 4(1)(2)}}{2(1)} & \left[\because m = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \right] \\
&= \frac{-2 \pm i2}{2} = -1 \pm i
\end{aligned}$$

$$\text{Particular Integral} = \frac{1}{D^2 + 2D + 2} (\cos x + x^2 + 7)$$

$$P.I_1 = \frac{1}{D^2 + 2D + 2} \cos x$$

$$= \frac{1}{-1+2D+2} \cos x \quad (\text{Replacing } D^2 \text{ by } -1)$$

$$= \frac{\cos x}{1+2D} \times \frac{1-2D}{1-2D}$$

$$= \frac{\cos x - 2D(\cos x)}{1-4D^2}$$

$$= \frac{\cos x + 2 \sin x}{1-4(-1)} = \frac{\cos x + 2 \sin x}{5}$$

$$P.I_2 = \frac{1}{D^2 + 2D + 2} (x^2 + 7)$$

$$= \frac{1}{2 \left[1 + \left(\frac{2D + D^2}{2} \right) \right]} (x^2 + 7)$$

$$= \frac{1}{2} \left[1 + \left(\frac{2D + D^2}{2} \right) \right]^{-1} (x^2 + 7)$$

$$= \frac{1}{2} \left[1 - \left(\frac{2D + D^2}{2} \right) + \left(\frac{2D + D^2}{2} \right)^2 - \dots \right] (x^2 + 7)$$

$$= \frac{1}{2} \left[1 - D - \frac{D^2}{2} + D^2 \right] (x^2 + 7) \quad [\because D^3(x^2 + 7) = 0]$$

$$= \frac{1}{2} [x^2 + 7 - 2x - 1 + 2]$$

$$P.I = \frac{1}{2} [x^2 - 2x + 8]$$

$\therefore$ The complete solution is $y = C.F + P.I$

$$y = e^{-x} (A \cos x + B \sin x) + \frac{\cos x + 2 \sin x}{5} + \frac{1}{2} (x^2 - 2x + 8)$$

Example 8: Solve $(D^2 + D + 2)y = e^{2x} + \sin 3x + 3x^2$

Solution:

The given differential equation is $(D^2 + D + 2)y = e^{2x} + \sin 3x + 3x^2$

The auxiliary equation is $m^2 + m + 2 = 0$

$$m = \frac{-1 \pm \sqrt{1-4}}{2} = -\frac{1}{2} \pm \frac{i\sqrt{3}}{2}$$

$$\therefore C.F. = e^{\frac{1}{2}x} \left[A \cos \frac{\sqrt{3}}{2}x + B \sin \frac{\sqrt{3}}{2}x \right]$$

$$\text{Particular Integral} = \frac{1}{D^2 + D + 2} (e^{2x} + \sin 3x + 3x^2)$$

$$P.I._1 = \frac{1}{D^2 + D + 2} e^{2x}$$

$$= \frac{1}{4 + 2 + 2} e^{2x} \quad (\text{Replacing } D \text{ by } 2)$$

$$P.I._1 = \frac{e^{2x}}{8}$$

$$P.I._2 = \frac{1}{D^2 + D + 2} \sin 3x \quad (\text{Replacing } D^2 \text{ by } -3^2)$$

$$= \frac{\sin 3x}{D-7} \times \frac{D+7}{D+7}$$

$$= \frac{D(\sin 3x) + 7 \sin 3x}{D^2 - 49}$$

$$= \frac{3 \cos 3x + 7 \sin 3x}{-9 - 49}$$

$$P.I._2 = -\frac{1}{58} (3 \cos 3x + 7 \sin 3x)$$

$$\begin{aligned}
P.I_3 &= \frac{1}{D^2 + D + 2} 3x^2 \\
&= \frac{1}{2 \left(1 + \frac{D + D^2}{2} \right)} 3x^2 = \frac{1}{2} \left(1 + \frac{D + D^2}{2} \right)^{-1} 3x^2 \\
&= \frac{1}{2} \left(1 - \left(\frac{D + D^2}{2} \right) + \left(\frac{D + D^2}{2} \right)^2 \right) 3x^2 \quad [\because D^3(3x^2) = 0] \\
&= \frac{1}{2} \left(1 - \left(\frac{D}{2} \right) - \frac{D^2}{2} + \left(\frac{D^2}{4} \right) \right) 3x^2 \\
&= \frac{1}{2} \left(3x^2 - \left(\frac{D(3x^2)}{2} \right) - \frac{D^2(3x^2)}{2} + \left(\frac{D^2(3x^2)}{4} \right) \right) \\
&= \frac{1}{2} \left(3x^2 - \left(\frac{6x}{2} \right) - \frac{6}{2} + \left(\frac{6}{4} \right) \right) = \frac{1}{2} \left(3x^2 - 3x - \frac{6}{4} \right) \\
&= \frac{3}{2} \left(x^2 - x - \frac{1}{2} \right) \\
P.I_3 &= \frac{3}{2} \left(x^2 - x - \frac{1}{2} \right)
\end{aligned}$$

$\therefore$ The complete solution is $y = C.F + P.I_1 + P.I_2 + P.I_3$

$$y = e^{-\frac{1}{2}x} \left[A \cos \frac{\sqrt{3}}{2} x + B \sin \frac{\sqrt{3}}{2} x \right] + \frac{e^{2x}}{8} - \frac{1}{58} (3 \cos 3x + 7 \sin 3x) + \frac{3}{2} \left(x^2 - x - \frac{1}{2} \right)$$

Example 9: Solve $(D^3 + 3D^2 + 2D)y = x^2$

Solution:

The given differential equation is $(D^3 + 3D^2 + 2D)y = x^2$

The auxiliary equation is $m^3 + 3m^2 + 2m = 0$

$$m(m^2 + 3m + 2) = 0$$

$$m(m+1)(m+2) = 0$$

$$\therefore m_1 = 0, \quad m_2 = -1, \quad m_3 = -2$$

$$\therefore C.F = A + Be^{-x} + Ce^{-2x}$$

$$\begin{aligned} \text{Particular Integral} &= \frac{1}{2D + 3D^2 + D^3} x^2 \\ &= \frac{1}{2D \left[1 + \left(\frac{3D^2 + D^3}{2D} \right) \right]} x^2 \\ &= \frac{1}{2D \left[1 + \left(\frac{3D + D^2}{2} \right) \right]} x^2 \\ &= \frac{1}{2D} \left[1 + \left(\frac{3D + D^2}{2} \right) \right]^{-1} x^2 \\ &= \frac{1}{2D} \left[1 - \left(\frac{3D + D^2}{2} \right) + \left(\frac{3D + D^2}{2} \right)^2 - \dots \right] x^2 \\ &= \frac{1}{2D} \left[1 - \frac{3D}{2} - \frac{D^2}{2} + \frac{9D^2}{4} \right] x^2 \quad \left[\because D^3(x^2) = 0 \right] \\ &= \frac{1}{2D} \left[1 - \frac{3D}{2} + \frac{7D^2}{4} \right] x^2 \\ &= \frac{1}{2D} \left[x^2 - 3x + \frac{7}{2} \right] = \frac{1}{2} \int \left(x^2 - 3x + \frac{7}{2} \right) dx \\ &= \frac{1}{2} \left[\frac{x^3}{3} - \frac{3x^2}{2} + \frac{7x}{2} \right] \end{aligned}$$

$$P.I. = \frac{x^3}{6} - \frac{3x^2}{4} + \frac{7x}{4}$$

The complete solution is $y = C.F. + P.I.$

$$y = A + Be^{-x} + Ce^{-2x} + \frac{x^3}{6} - \frac{3x^2}{4} + \frac{7x}{4}$$

Example 10: Solve $(D^2 - D + 1)y = 1 - 3x^2 + x^3$

(Nov- 2011)

Solution:

The given differential equation is $(D^2 - D + 1)y = 1 - 3x^2 + x^3$

The auxiliary equation is $m^2 - m + 1 = 0$

$$m = \frac{1 \pm \sqrt{1-4}}{2} = \frac{1 \pm i\sqrt{3}}{2}$$

$$\therefore C.F. = e^{\frac{1}{2}x} \left(A \cos \frac{\sqrt{3}}{2} x + B \sin \frac{\sqrt{3}}{2} x \right)$$

$$\begin{aligned} \text{Particular Integral} &= \frac{1}{D^2 - D + 1} (x^3 - 3x^2 + 1) \\ &= \frac{1}{1 - (D - D^2)} \cdot (x^3 - 3x^2 + 1) \\ &= [1 - (D - D^2)]^{-1} (x^3 - 3x^2 + 1) \\ &= [1 - (D - D^2) + (D - D^2)^2 + (D - D^2)^3 + \dots] (x^3 - 3x^2 + 1) \\ &= [1 + D - D^2 + D^2 - 2D^3 + D^3] (x^3 - 3x^2 + 1) \\ &\quad (\because D^4(x^3 - 3x^2 + 1) = 0) \\ &= x^3 - 3x^2 + 1 + 3x^2 - 6x - 2(6) + 6 \\ &= x^3 - 6x - 5 \end{aligned}$$

$\therefore$ The complete solution is $y = C.F. + P.I.$

$$y = e^{\frac{1}{2}x} \left(A \cos \frac{\sqrt{3}}{2} x + B \sin \frac{\sqrt{3}}{2} x \right) + x^3 - 6x - 5$$

Exercise

Solve the following equations

$$1. \frac{d^2 y}{dx^2} - 4y = x^2$$

$$2. (D^2 - 4D + 3)y = x^2$$

$$3. (D^3 - D^2 - D + 1)y = 1 + x^2$$

$$4. (D^2 - 1)y = 2 + 5x$$

$$5. (D^2 + D + 1)y = x^2$$

$$6. (D^3 - 3D^2 - D + 8)y = x$$

$$7. (D^2 - 6D + 9)y = x^2 + 3x + 4$$

$$8. (D^2 - 4D + 4)y = x^3$$

$$9. (D^3 + 8)y = x^4 + 2x + 1$$

$$10. (D^2 - 3D + 2)y = 2x^2 + 1$$

Answers

$$1. y = Ae^{2x} + Be^{-2x} - \frac{1}{4} \left(x^2 + \frac{1}{2} \right)$$

$$2. y = Ae^{3x} + Be^x + \frac{x^2}{3} + \frac{8x}{9} + \frac{26}{27}$$

$$3. y = Ae^{-x} + e^x + (B + Cx) + x^2 + 2x + 5$$

$$4. y = Ae^{-x} + Be^x - 2 - 5x$$

$$5. y = e^{-\frac{x}{2}} \left(A \cos \frac{\sqrt{3}}{2} x + B \sin \frac{\sqrt{3}}{2} x \right) x^2 - 2x$$

$$6. y = Ae^x + Be^{-2x} + Ce^{4x} + \frac{1}{8} \left(x + \frac{3}{4} \right)$$

$$7. y = (A + Bx)e^{3x} + \frac{1}{9} \left(x^2 + \frac{13}{3} + \frac{30}{3} \right)$$

$$8. y = (A + Bx)e^{3x} + \frac{1}{8} (2x^3 + 6x^2 + 9x + 6)$$

$$9. y = Ae^{-2x} + e^x + \left(B \cos \sqrt{3}x + C \sin \sqrt{3}x \right) + \frac{1}{8} (x^4 - x + 1)$$

$$10. y = Ae^x + Be^{2x} + x^2 + 3x + 4$$

Type IV

If $f(x) = e^{ax} X$, where X is some function of x , then, $P.I = \frac{1}{f(D)} e^{ax} X$
 $= e^{ax} \frac{1}{f(D+a)} X$. (i.e.) e^{ax} which is on the right of $\frac{1}{f(D)}$ may be taken out to the left provided D is replaced by $D + a$.

Example 1: Solve $(D^2 - 4D + 13)y = e^{2x} \cos 3x$

Solution:

The given differential equation is $(D^2 - 4D + 13)y = e^{2x} \cos 3x$

The auxiliary equation is $m^2 - 4m + 13 = 0$

$$m = \frac{4 \pm \sqrt{16 - 52}}{2} = 2 \pm 3i$$

$$\therefore C.F. = e^{2x} (A \cos 3x + B \sin 3x)$$

$$\text{Particular Integral} = \frac{1}{D^2 - 4D + 13} e^{2x} \cos 3x$$

$$= e^{2x} \frac{1}{(D+2)^2 - 4(D+2) + 13} \cos 3x$$

$$= e^{2x} \frac{1}{D^2 + 4D + 4 - 4D - 8 + 13} \cos 3x$$

$$= e^{2x} \frac{1}{D^2 + 9} \cos 3x$$

$$= e^{2x} \frac{1}{-9 + 9} \cos 3x$$

$$= x e^{2x} \frac{1}{2D} (\cos 3x)$$

$$= \frac{x e^{2x}}{2} \int \cos 3x dx = \frac{x e^{2x}}{2} \times \frac{\sin 3x}{3}$$

$$P.I = \frac{xe^{2x} \sin 3x}{6}$$

∴ The complete solution is $y = C.F + P.I$

$$y = e^{2x} (A \cos 3x + B \sin 3x) + \frac{xe^{2x} \sin 3x}{6}$$

Example 2: Solve $y'' + y = e^{-x} \sin x$

Solution:

The given differential equation is $(D^2 + 1)y = e^{-x} \sin x$

The auxiliary equation is $(m^2 + 1) = 0$

$$m^2 = -1, m = \pm i$$

$$\therefore C.F = A \cos x + B \sin x$$

$$\text{Particular Integral } P.I = \frac{1}{D^2 + 1} e^{-x} \sin x$$

$$= e^{-x} \frac{1}{(D-1)^2 + 1} \sin x = e^{-x} \frac{1}{D^2 - 2D + 2} \sin x$$

$$= e^{-x} \frac{1}{-1 - 2D + 2} \sin x$$

$$= e^{-x} \frac{\sin x}{1 - 2D} \times \frac{1 + 2D}{1 + 2D}$$

$$= e^{-x} \frac{\sin x + 2D(\sin x)}{1 - 4D^2}$$

$$= e^{-x} \frac{\sin x + 2 \cos x}{1 - 4(-1)} = e^{-x} \frac{\sin x + 2 \cos x}{5}$$

∴ The complete solution is $y = C.F + P.I$

$$y = A \cos x + B \sin x + \frac{e^{-x}}{5} (\sin x + 2 \cos x)$$

Example 3: Solve $(D^2 + 4D + 4)y = e^{-2x} \sin x$ **(Apr- 2010)**

Solution:

The given differential equation is $(D^2 + 4D + 4)y = e^{-2x} \sin x$

The auxiliary equation is $m^2 + 4m + 4 = 0$

$$(m + 2)(m + 2) = 0$$

$$m = -2, -2$$

$$\therefore C.F. = e^{-2x}(Ax + B)$$

$$\text{Particular Integral} = \frac{1}{D^2 + 4D + 4} e^{-2x} \sin x$$

$$= e^{-2x} \frac{1}{(D - 2)^2 + 4(D - 2) + 4} \sin x$$

$$= e^{-2x} \frac{1}{D^2 - 4D + 4 + 4D - 8 + 4} \sin x$$

$$= e^{-2x} \frac{1}{D^2} \sin x = e^{-2x} \frac{\sin x}{-1}$$

$\therefore$ The complete solution is $y = C.F. + P.I$

$$y = e^{-2x}(Ax + B) - e^{-2x} \sin x$$

Example 4:

Solve $\frac{d^2 y}{dx^2} - 2 \frac{dy}{dx} + 4y = e^x \cos x + e^{2x}$ **(Jan- 2011)**

Solution:

The given differential equation is $(D^2 - 2D + 4)y = e^x \cos x + e^{2x}$

The auxiliary equation is $m^2 - 2D + 4 = 0$

$$m = \frac{2 \pm \sqrt{4 - 4(4)}}{2} = \frac{2 \pm \sqrt{-12}}{2} = \frac{2 \pm 2i\sqrt{3}}{2}$$

$$m = 1 \pm i\sqrt{3}$$

$$\therefore C.F. = e^x (A \cos \sqrt{3}x + B \sin \sqrt{3}x)$$

$$\text{Particular Integral} = \frac{1}{D^2 - 2D + 4} (e^x \cos x + e^{2x})$$

$$\begin{aligned} P.I_1 &= \frac{1}{D^2 - 2D + 4} e^x \cos x \\ &= e^x \frac{1}{(D+1)^2 - 2(D+1) + 4} \cos x \\ &= e^x \frac{1}{D^2 + 2D + 1 - 2D - 2 + 4} \cos x \\ &= e^x \frac{1}{D^2 + 3} \cos x \\ &= e^x \frac{1}{-1 + 3} \cos x \end{aligned}$$

$$P.I_1 = \frac{e^x \cos x}{2}$$

$$\begin{aligned} P.I_2 &= \frac{1}{D^2 - 2D + 4} e^{2x} \\ &= \frac{1}{4 - 2(2) + 4} e^{2x} \end{aligned}$$

$$P.I_2 = \frac{e^{2x}}{4}$$

$\therefore$ The complete solution is $y = C.F. + P.I$

$$y = e^x (A \cos \sqrt{3}x + B \sin \sqrt{3}x) + \frac{e^x \cos x}{2} + \frac{e^{2x}}{4}$$

Example 5: Solve $(D^2 + 1)y = x \sinh x$

Solution:

The given differential equation is $(D^2 + 1)y = x \sinh x$

The auxiliary equation is $m^2 + 1 = 0$

$$m^2 = -1$$

$$\therefore m = \pm i$$

$\therefore$ Complementary function C.F. = $A \cos x + B \sin x$

$$\text{Particular integral} = \frac{1}{(D^2 + 1)} x \sinh x$$

$$= \frac{1}{(D^2 + 1)} x \left(\frac{e^x - e^{-x}}{2} \right) \quad \left(\because \sinh x = \frac{e^x - e^{-x}}{2} \right)$$

$$= \frac{1}{2} \left[\frac{1}{(D^2 + 1)} x e^x - \frac{1}{(D^2 + 1)} x e^{-x} \right]$$

$$\therefore \text{P.I.} = \frac{1}{2} [P.I_1 - P.I_2] \quad \dots(1)$$

$$P.I_1 = \frac{1}{(D^2 + 1)} x e^x$$

$$= e^x \frac{1}{(D+1)^2 + 1} x \quad (\text{Replacing } D \text{ by } D+1)$$

$$= e^x \frac{1}{D^2 + 2D + 2} x$$

$$= e^x \frac{1}{2 \left[1 + \left(\frac{2D + D^2}{2} \right) \right]} x = \frac{e^x}{2} \left[1 + \left(\frac{2D + D^2}{2} \right) \right]^{-1} x$$

$$= \frac{e^x}{2} \left[1 - \left(\frac{2D + D^2}{2} \right) \right] x$$

$$\therefore P.I_1 = \frac{e^x}{2}(x-1)$$

$$P.I_2 = \frac{1}{(D^2+1)}xe^{-x}$$

$$= e^{-x} \frac{1}{(D-1)^2+1} x \quad (\text{Replacing } D \text{ by } D-1)$$

$$= e^{-x} \frac{1}{D^2-2D+2} x = e^{-x} \frac{1}{2 \left[1 - \left(\frac{2D-D^2}{2} \right) \right]} x$$

$$= \frac{e^{-x}}{2} \left[1 - \left(\frac{2D-D^2}{2} \right) \right]^{-1} x = \frac{e^{-x}}{2} \left[1 + \left(\frac{2D-D^2}{2} \right) \right] x$$

$$\therefore P.I_2 = \frac{e^{-x}}{2}(x+1)$$

$$\begin{aligned} \text{From Equation (1), P.I.} &= \frac{1}{2} [P.I_1 - P.I_2] = \frac{1}{2} \left[\frac{e^x}{2}(x-1) - \frac{e^{-x}}{2}(x+1) \right] \\ &= \frac{1}{2} \left[\frac{xe^x - e^x - xe^{-x} - e^{-x}}{2} \right] \\ &= \frac{1}{2} [x \sinh x - \cosh x] \end{aligned}$$

The Complete Solution is $y = \text{C.F.} + \text{P.I.}$

$$y = A \cos x + B \sin x + \frac{1}{2} [x \sinh x - \cosh x]$$

Example 6: Solve $(D^3 - 7D - 6)y = e^{2x}(1+x)$ **(Nov-2008)**

Solution:

The given differential equation is $(D^3 - 7D - 6)y = e^{2x}(1+x)$

The auxiliary equation is $m^2 - 7m - 6 = 0$

$$\begin{array}{c|cccc} 3 & 1 & 0 & -7 & -6 \\ & 0 & 3 & 9 & 6 \\ \hline & 1 & 3 & 2 & 0 \end{array}$$

$\therefore m_1 = 3$ is a root

The reduced equation is $m^2 + 3m + 2 = 0$

$$m_2 = -1, m_3 = -2$$

$\therefore$ Complementary function C.F. = $Ae^{-x} + Be^{-2x} + Ce^{3x}$

$$\begin{aligned} \text{Particular integral} &= \frac{1}{D^3 - 7D - 6} e^{2x}(1+x) \\ &= e^{2x} \frac{1}{(D+2)^3 - 7(D+2) - 6} (1+x) && \text{(Replacing } D \text{ by } D+2) \\ &= e^{2x} \frac{1}{D^3 + 6D^2 + 5D - 12} (1+x) \\ &= e^{2x} \frac{1}{-12 \left[1 - \left(\frac{5D + 6D^2 + D^3}{12} \right) \right]} (1+x) \\ &= -\frac{e^{2x}}{12} \left[1 - \left(\frac{5D + 6D^2 + D^3}{12} \right) \right]^{-1} (1+x) \\ &= -\frac{e^{2x}}{12} \left[1 + \frac{5D}{12} \right] (1+x) = -\frac{e^{2x}}{12} \left[1 + x + \frac{5}{12} \right] \\ &= -\frac{e^{2x}}{12} \left[x + \frac{17}{2} \right] \end{aligned}$$

The Complete Solution is $Y = \text{C.F.} + \text{P.I.}$

$$y = Ae^{-x} + Be^{-2x} + Ce^{3x} - \frac{e^{2x}}{12} \left[x + \frac{17}{2} \right]$$

Example 7: Solve $(D^2 + 2D + 1)y = (x + e^x)^2$ **(May-2011)**

Solution:

The given differential equation is $(D^2 + 2D + 1)y = (x + e^x)^2$

The auxiliary equation is $m^2 + 2m + 1 = 0$

$$(m+1)(m+1) = 0 \Rightarrow m = -1, -1$$

Complementary function C.F. = $e^{-x}(Ax + B)$

$$\text{Particular integral} = \frac{1}{(D^2 + 2D + 1)}(x + e^x)^2$$

$$= \frac{1}{(D^2 + 2D + 1)}(x^2 + e^{2x} + 2xe^x)$$

$$P.I_1 = \frac{1}{(D^2 + 2D + 1)}x^2$$

$$= \frac{1}{(1 + (2D + D^2))}x^2 = [1 + (2D + D^2)]^{-1}x^2$$

$$= [1 - (2D + D^2) + (2D + D^2)^2 - \dots]x^2$$

$$= [1 - 2D - D^2 + 4D^2]x^2$$

$$= x^2 - 4x - 2 + 8 = x^2 - 4x + 6$$

$$P.I_2 = \frac{1}{(D^2 + 2D + 1)}2xe^x$$

$$= 2e^x \frac{1}{((D+1)^2 + 2(D+1) + 1)}x \quad (\text{Replacing } D \text{ by } D+1)$$

$$\begin{aligned}
&= 2e^x \frac{1}{(D^2 + 4D + 4)} x = 2e^x \frac{1}{4 \left[1 + \left(D + \frac{D^2}{4} \right) \right]} x \\
&= \frac{e^x}{2} \left(1 + \left(D + \frac{D^2}{4} \right) \right)^{-1} x = \frac{e^x}{2} (1 - D) x \\
\therefore P.I_2 &= \frac{e^x}{2} (x - 1) \\
P.I_3 &= \frac{1}{(D^2 + 2D + 1)} e^{2x} \\
&= \frac{1}{(4 + 4 + 1)} e^{2x} \quad (\text{Replacing } D \text{ by } 2) \\
&= \frac{e^{2x}}{9}
\end{aligned}$$

The Complete Solution is $y = \text{C.F.} + \text{P.I.}$

$$y = e^{-x} (Ax + B) + x^2 - 4x + 6 + \frac{e^x}{2} (x - 1) + \frac{e^{2x}}{9}$$

Example 8: Solve $(D^2 - 2D + 2)y = e^x x^2 + 5 + e^{-2x}$ (Jan -2013)

Solution:

The given differential equation is $(D^2 - 2D + 2)y = e^x x^2 + 5 + e^{-2x}$

The auxiliary equation is $m^2 - 2m + 2 = 0$

$$m = \frac{2 \pm \sqrt{4 - 8}}{2} = \frac{2 \pm \sqrt{-4}}{2} = 1 \pm i$$

Complementary function C.F. = $e^x (A \cos x + B \sin x)$

$$\text{Particular integral} = \frac{1}{(D^2 - 2D + 2)} (e^x x^2 + 5 + e^{-2x})$$

$$\begin{aligned}
P.I_1 &= \frac{1}{(D^2 - 2D + 2)} e^x x^2 \\
&= e^x \frac{1}{((D+1)^2 - 2(D+1) + 2)} x^2 \\
&= e^x \frac{1}{(D^2 + 2D + 1 - 2D - 2 + 2)} x^2 = e^x \frac{1}{(D^2 + 1)} x^2 \\
&= e^x (1 + D^2)^{-1} x^2 \\
&= e^x (1 - D^2 + D^4 \dots) x^2 = e^x (1 - D^2) x^2 = e^x (x^2 - 2) \\
P.I_2 &= \frac{1}{(D^2 - 2D + 2)} 5e^{0x} = \frac{5}{2} \\
P.I_3 &= \frac{1}{(D^2 - 2D + 2)} e^{-2x} \\
&= \frac{1}{(4 + 4 + 2)} e^{-2x} \quad \text{(Replacing D by -2)} \\
&= \frac{e^{-2x}}{10}
\end{aligned}$$

The Complete Solution is $y = \text{C.F.} + \text{P.I.}$

$$y = e^x (A \cos x + B \sin x) + e^x (x^2 - 2) + \frac{5}{2} + \frac{e^{-2x}}{10}$$

Example 9: Solve $(D^2 + 2D - 3)y = e^x \cos x + e^{-x} x^2$ **(Nov -2006)**

Solution:

The given differential equation is $(D^2 + 2D - 3)y = e^x \cos x + e^{-x} x^2$

The auxiliary equation is $m^2 + 2m - 3 = 0$

$$(m-1)(m+3) = 0$$

$$m_1 = 1, m_2 = -3$$

$$\text{Complementary function} = Ae^x + Be^{-3x}$$

$$\text{Particular integral} = \frac{1}{(D^2 + 2D - 3)}(e^x \cos x + e^{-x} x^2)$$

$$\begin{aligned} P.I_1 &= \frac{1}{(D^2 + 2D - 3)}(e^x \cos x) \\ &= e^x \frac{1}{((D+1)^2 + 2(D+1) - 3)}(\cos x) \quad (\text{Replacing } D \text{ by } D+1) \\ &= e^x \frac{1}{(D^2 + 2D + 1 + 2D + 2 - 3)}(\cos x) = e^x \frac{1}{(D^2 + 4D)}(\cos x) \\ &= e^x \frac{1}{(-1 + 4D)}(\cos x) \quad (\text{Replacing } D^2 \text{ by } -1) \\ &= e^x \frac{(\cos x)}{(-1 + 4D)} \times \frac{(4D + 1)}{(4D + 1)} = e^x \frac{4D(\cos x) + \cos x}{(16D^2 - 1)} \\ &= e^x \frac{4(-\sin x) + \cos x}{(-17)} \quad (\text{Replacing } D^2 \text{ by } -1) \\ &= \frac{e^x}{17}[4\sin x - \cos x] \end{aligned}$$

$$\begin{aligned} P.I_2 &= \frac{1}{(D^2 + 2D - 3)}(e^{-x} x^2) \\ &= e^{-x} \frac{1}{((D-1)^2 + 2(D-1) - 3)}(x^2) \\ &= e^{-x} \frac{1}{(D^2 - 2D + 1 + 2D - 2 - 3)}(x^2) = e^{-x} \frac{1}{(D^2 - 4)}(x^2) \\ &= e^{-x} \frac{1}{-4\left(1 - \frac{D^2}{4}\right)}(x^2) = \frac{e^{-x}}{-4}\left(1 - \frac{D^2}{4}\right)^{-1}(x^2) \end{aligned}$$

$$= \frac{e^{-x}}{-4} \left(1 + \frac{D^2}{4} + \left(\frac{D^2}{4} \right)^2 + \dots \right) (x^2) = \frac{e^{-x}}{-4} \left(x^2 + \frac{1}{2} \right)$$

The Complete Solution is $y = \text{C.F.} + \text{P.I.}$

$$y = Ae^x + Be^{-3x} + \frac{e^x}{17} [4 \sin x - \cos x] - \frac{e^{-x}}{4} \left(x^2 + \frac{1}{2} \right)$$

Exercise

Solve the following equations

1. $(D^2 + 3D + 2)y = xe^x$
2. $(D^2 + 2D + 5)y = e^{-x} \sin 2x + x$
3. $(D^2 - 4D + 4)y = 2e^{2x} + xe^{2x}$
4. $(D^3 - 3D^2 - 6D + 8)y = x^2 e^{4x}$
5. $(D - 1)^3 y = xe^x + e^x$
6. $(D^2 - 1)y = xe^x$
7. $(D^2 + 3D + 4)y = x \sinh x$
8. $(D^2 + 2D - 3)y = e^{2x}(1 + x^2)$
9. $(D^3 - 8)y = xe^x + 3e^{-x}$
10. $(D^2 + 4D + 3)y = e^{-x} \sin x + xe^{3x}$

Answers

1. $y = Ae^{-x} + Be^{-2x} + \frac{e^x}{36}(6x - 5)$
2. $y = e^{-x}(A \cos 2x + B \sin 2x) - \frac{xe^{-x}}{4} \cos 2x + \frac{1}{25}(5x - 2)$
3. $y = e^{2x}(A + Bx) + x^2 e^{2x} + \frac{x^3 e^{2x}}{6}$
4. $y = Ae^{-2x} + Be^x + Ce^{4x} + \frac{e^{4x}}{324}(6x^2 - 9x + 7)$
5. $y = e^x(Ax^2 + Bx + C) + e^x \frac{x^3}{24}(x + 4)$
6. $y = Ae^x + Be^{-x} + \frac{e^x(x^2 - x)}{4}$

$$7. \quad y = e^{-\frac{3}{2}x} \left(A \cos \frac{\sqrt{7}}{2} x + B \sin \frac{\sqrt{7}}{2} x \right) + \frac{e^x}{128} (8x - 5) - \frac{e^{-x}}{8} (2x - 1)$$

$$8. \quad y = Ae^{-2x} + Be^x + \frac{e^{2x}}{5} \left(x^2 - \frac{12x}{5} + \frac{62}{25} \right)$$

$$9. \quad y = Ae^{2x} + (B \cos \sqrt{3}x + C \sin \sqrt{3}x)e^{-x} - \frac{e^x}{7} \left(x + \frac{3}{7} \right) - \frac{e^{-x}}{3}$$

$$10. \quad y = Ae^{-3x} + Be^{-x} - \frac{e^{-x}}{5} (2 \cos x + \sin x) - e^{-3x} \left(\frac{x^2 + x}{4} \right)$$

Type V

If $f(x) = x^n \sin ax$ or $x^n \cos ax$, then

$$\text{P.I.} = \frac{1}{f(D)} x^n \sin ax = \text{Imaginary part of } e^{iax} \frac{1}{f(D+ia)} x^n \text{ and}$$

$$\text{P. I.} = \frac{1}{f(D)} x^n \cos ax = \text{Real part of } e^{iax} \frac{1}{f(D+ia)} x^n$$

Example 1: Solve $(D^2 - 2D + 1)y = x \sin x$

Solution:

The given differential equation is $(D^2 - 2D + 1)y = x \sin x$

The auxiliary equation is $m^2 - 2m + 1 = 0$

$$(m-1)(m-1) = 0$$

$$m = 1(\text{twice})$$

Complementary function = $(Ax + B)e^x$

$$\text{Particular integral} = \frac{1}{(D^2 - 2D + 1)} x \sin x$$

$$= \text{I.P.of } \frac{1}{(D^2 - 2D + 1)} x e^{ix} \quad (\because e^{ix} = \cos x + i \sin x)$$

$$= \text{I.P.of } e^{ix} \frac{1}{((D+i)^2 - 2(D+i) + 1)} x \quad (\text{Replacing } D \text{ by } D+i)$$

$$= \text{I.P.of } e^{ix} \frac{1}{(D^2 - 2(1-i)D - 2i)} x$$

$$= \text{I.P.of } e^{ix} \frac{1}{-2i \left(1 + \frac{D^2 - 2(1-i)D}{-2i} \right)} x$$

$$= \text{I.P.of } e^{ix} \frac{1}{-2i \left(1 - \frac{D^2 - 2(1-i)D}{2i} \right)} x$$

$$= \text{I.P.of } \frac{e^{ix}}{-2i} \left(1 - \frac{D^2 - 2(1-i)D}{2i} \right)^{-1} x$$

$$= \text{I.P.of } \frac{e^{ix}}{-2i} \left(1 + \frac{D^2 - 2(1-i)D}{2i} + \left(\frac{D^2 - 2(1-i)D}{2i} \right)^2 + \dots \right) x$$

$$= \text{I.P.of } e^{ix} \frac{i}{2} \left(1 + \frac{D^2 - 2(1-i)D}{2i} + \left(\frac{D^2 - 2(1-i)D}{2i} \right)^2 + \dots \right) x$$

$$= \text{I.P.of } e^{ix} \frac{i}{2} (x+1+i)$$

$$= \text{I.P.of } (\cos x + i \sin x) \left(\frac{i}{2} (x+1+i) \right)$$

$$= \text{I.P.of } \frac{1}{2} (i \cos x - \sin x) ((x+1+i))$$

$$= \frac{1}{2}x \cos x + \frac{1}{2} \cos x - \frac{1}{2} \sin x$$

The Complete Solution is $y = \text{C.F.} + \text{P.I.}$

$$y = (Ax + B)e^x + \frac{1}{2}x \cos x + \frac{1}{2} \cos x - \frac{1}{2} \sin x$$

Example 2: Solve $(D^2 - 2D + 1)y = xe^x \sin x$

Solution:

The given differential equation is $(D^2 - 2D + 1)y = xe^x \sin x$

The auxiliary equation is $m^2 - 2m + 1 = 0$

$$(m-1)(m-1) = 0$$

$$m = 1(\text{twice})$$

Complementary function = $(Ax + B)e^x$

Particular integral = $\frac{1}{(D^2 - 2D + 1)} xe^x \sin x$

$$= e^x \frac{1}{((D+1)^2 - 2(D+1) + 1)} x \sin x \text{ (Replacing } D \text{ by } D+1)$$

$$= e^x \frac{1}{(D^2 + 1 + 2D - 2D - 2 + 1)} x \sin x$$

$$= e^x \frac{1}{(D^2)} x \sin x = e^x \frac{1}{D} (-x \cos x + \sin x)$$

$$= e^x (-x \sin x - \cos x - \cos x)$$

$$= -e^x (x \sin x + 2 \cos x)$$

The Complete Solution is $y = \text{C.F.} + \text{P.I.}$

$$y = (Ax + B)e^x - e^x (x \sin x + 2 \cos x)$$

Example 3: Solve $(D^2 - 1)y = x^2 \cos x$

Solution:

The given differential equation is $(D^2 - 1)y = x^2 \cos x$

The auxiliary equation is $m^2 - 1 = 0$

$$m^2 = 1$$

$$m_1 = 1, m_2 = -1$$

Complementary function = $Ae^x + Be^{-x}$

$$\text{Particular integral} = \frac{1}{(D^2 - 1)} x^2 \cos x$$

$$= \text{Real part of } \frac{1}{(D^2 - 1)} x^2 e^{ix} \quad (\because e^{ix} = \cos x + i \sin x)$$

$$= \text{R.P of } e^{ix} \frac{1}{((D+i)^2 - 1)} x^2 = \text{R.P of } e^{ix} \frac{1}{(D^2 + 2iD - 1 - 1)} x^2$$

$$= \text{R.P of } e^{ix} \frac{1}{-2 \left(1 - \frac{(D^2 + 2iD)}{2} \right)} x^2 = \text{R.P of } \frac{e^{ix}}{-2} \left(1 - \frac{(D^2 + 2iD)}{2} \right)^{-1} x^2$$

$$= \text{R.P of } \frac{e^{ix}}{-2} \left(1 + \frac{(D^2 + 2iD)}{2} + \left(\frac{(D^2 + 2iD)}{2} \right)^2 + \dots \right) x^2$$

$$= \text{R.P of } \frac{e^{ix}}{-2} \left(1 + iD + \frac{D^2}{2} - D^2 \right) x^2 = \text{R.P of } \frac{e^{ix}}{-2} (x^2 + 2ix + 1 - 2)$$

$$= \text{R.P of } \left[\frac{1}{-2} (\cos x + i \sin x) (x^2 + 2ix - 1) \right]$$

$$= \left[-\frac{1}{2} (x^2 \cos x - \cos x - 2x \sin x) \right]$$

$$\text{Particular integral} = x \sin x + \frac{1}{2} (1 - x^2) \cos x$$

The Complete Solution is $y = \text{C.F.} + \text{P.I.}$

$$y = Ae^x + Be^{-x} + x \sin x + \frac{1}{2}(1-x^2)\cos x$$

Example 4: Solve $(D^2 + 4)y = x^2 \cos 2x$

Solution:

The given differential equation is $(D^2 + 4)y = x^2 \cos 2x$

The auxiliary equation is $m^2 + 4 = 0$

$$m^2 = -4, m_1 = 2i, m_2 = -2i$$

Complementary function = $A \cos 2x + B \sin 2x$

$$\text{Particular integral} = \frac{1}{(D^2 + 4)} x^2 \cos 2x$$

$$= \text{Real part of } \frac{1}{(D^2 + 4)} x^2 e^{i2x}$$

$$= \text{R.P of } e^{i2x} \frac{1}{((D+2i)^2 + 4)} x^2 \quad (\text{Replacing D by D+2i})$$

$$= \text{R.P of } e^{i2x} \frac{1}{(D^2 + 4iD)} x^2 = \text{R.P of } e^{i2x} \left(\frac{1}{4iD \left(1 + \frac{D}{4i} \right)} \right) x^2$$

$$= \text{R.P of } \frac{e^{i2x}}{4iD} \left(1 + \frac{D}{4i} \right)^{-1} x^2$$

$$= \text{R.P of } \frac{e^{i2x}}{4iD} \left(1 - \frac{D}{4i} + \left(\frac{D}{4i} \right)^2 - \left(\frac{D}{4i} \right)^3 \dots \right) x^2$$

$$= \text{R.P of } \frac{e^{i2x}}{4i} \left(\frac{1}{D} + \frac{i}{4} - \frac{D}{16} - \frac{iD^2}{64} \right) x^2$$

$$= \text{R.P of } \frac{-ie^{i2x}}{4} \left(\frac{x^3}{3} + \frac{ix^2}{4} - \frac{x}{8} - \frac{i}{32} \right)$$

$$\begin{aligned}
&= \text{R.P of } \frac{1}{4}(\sin 2x - i \cos 2x) \left(\frac{x^3}{3} - \frac{x}{8} + \frac{i}{4} \left(x^2 - \frac{1}{8} \right) \right) \\
&= \frac{1}{4} \left[\left(\frac{x^3}{3} - \frac{x}{8} \right) \sin 2x + \frac{1}{4} \left(x^2 - \frac{1}{8} \right) \cos 2x \right]
\end{aligned}$$

The Complete Solution is $y = \text{C.F} + \text{P.I}$

$$\therefore y = A \cos 2x + B \sin 2x + \frac{1}{4} \left[\left(\frac{x^3}{3} - \frac{x}{8} \right) \sin 2x + \frac{1}{4} \left(x^2 - \frac{1}{8} \right) \cos 2x \right]$$

Exercise

Solve the following differential equations

1. $(D^2 - 1)y = x \sin 3x + \cos x$

4. $(D^3 - 1)y = x \sin x$

2. $(D^2 + 4)y = x \sin x$

5. $(D^2 - 4D + 4)y = 8x^2 e^{2x} \sin 2x$

3. $(D^4 + 2D^2 + 1)y = x^2 \cos x$

Answers

1. $y = Ae^x + Be^{-x} - \frac{1}{10} \left[\frac{3}{5} \cos 3x + x \sin 3x + 5 \cos 3x \right]$

2. $y = A \cos 2x + B \sin 2x + \frac{x \sin x}{3} - \frac{2}{9} \cos x$

3. $y = (A + Bx) \cos x + (C + Dx) \sin x + \frac{1}{12} x^3 \sin x - \frac{1}{48} (x^4 - 9x^2) \cos x$

4. $y = Ae^x + e^{-\frac{x}{2}} \left(B \cos \frac{\sqrt{3}}{2} x + C \sin \frac{\sqrt{3}}{2} x \right) + \frac{x}{2} (\cos x - \sin x) - \frac{3}{2} \cos x$

5. $y = (A + Bx) e^{2x} + e^{2x} \left[(3 - 2x^2) \sin 2x - 4x \cos 2x \right]$

Euler's Linear Equation of Higher Order with Variable coefficients

The equation of the form $a_0 x^n \frac{d^n y}{dx^n} + a_1 x^{n-1} \frac{d^{n-1} y}{dx^{n-1}} + a_2 x^{n-2} \frac{d^{n-2} y}{dx^{n-2}} + \dots + a_{n-1} x \frac{dy}{dx} + a_n y = X$

...(1), where $a_0, a_1, a_2, \dots, a_n$ are constants and X is a function of x is called Euler's homogeneous linear differential equation. In each term in the L.H.S. of equation (1), the power of x and the order of the derivative are equal. Hence, this equation is called as a homogeneous equation. Equation (1) can be reduced to a linear differential equation with constant coefficients by changing the independent variable from x to z by means of the

transformation $x = e^z$ (or) $z = \log x$, $\frac{dz}{dx} = \frac{1}{x}$

$$\frac{dy}{dx} = \frac{dy}{dz} \frac{dz}{dx} = \frac{1}{x} \frac{dy}{dz}$$

$$\therefore x \frac{dy}{dx} = \frac{dy}{dz} \quad \dots(2)$$

Differentiating both sides of (2) with respect to x ,

$$x \frac{d^2 y}{dx^2} + \frac{dy}{dx} = \frac{d^2 y}{dz^2} \frac{1}{x}$$

$$\text{(i.e.) } x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} = \frac{d^2 y}{dz^2}$$

$$\text{(i.e.) } x^2 \frac{d^2 y}{dx^2} = \frac{d^2 y}{dz^2} - \frac{dy}{dz} \quad (\because \text{by equation (2)}) \dots(3)$$

Denoting $\frac{d}{dx}$ by D and $\frac{d}{dz}$ by θ ,

$$\text{Equation (2)} \Rightarrow xD = \theta \quad \dots(4)$$

$$\text{and Equation (3)} \Rightarrow x^2 D^2 = \theta^2 - \theta = \theta(\theta - 1) \quad \dots(5)$$

$$\text{Similarly we can show that } x^3 D^3 = \theta(\theta - 1)(\theta - 2) \quad \dots(6)$$

$$x^4 D^4 = \theta(\theta - 1)(\theta - 2)(\theta - 3) \quad \dots(7) \text{ and so on.}$$

Substituting (4), (5), (6), (7) in equation (1), we get a differential equation with constant coefficient and can be solved by any one of the known methods.

Example 1: Solve $(x^2 D^2 - 4xD + 6)y = 0$

Solution:

The given differential equation is $(x^2 D^2 - 4xD + 6)y = 0$... (1)

Put $x = e^z$ (or) $z = \log x$

$$xD = \theta \quad \dots (2)$$

$$x^2 D^2 = \theta(\theta - 1) \quad \dots (3)$$

Substituting equation (2) and (3) in equation (1), we get

$$(\theta(\theta - 1) - 4\theta + 6)y = 0$$

$$(\theta^2 - 5\theta + 6)y = 0$$

The Auxiliary equation is $m^2 - 5m + 6 = 0$

$$(m - 3)(m - 2) = 0$$

$$m_1 = 2, m_2 = 3$$

$\therefore$ Complementary Function = $Ae^{2z} + Be^{3z}$

$$= Ae^{2 \log x} + Be^{3 \log x}$$

$$= Ae^{\log x^2} + Be^{\log x^3}$$

The Complete Solution is $y = \text{C.F.} + \text{P.I.}$

$$y = Ax^2 + Bx^3$$

Example 2: Solve $x^2 y'' - 3xy' + 4y = x^2$ (Nov- 2003)

Solution:

The given differential equation can be written in the operator form as

$$(x^2 D^2 - 3xD + 4)y = x^2 \quad \dots (1)$$

Put $x = e^z$ (or) $z = \log x$ and $e^{2z} = x^2$

$$xD = \theta \quad \dots(2)$$

$$x^2 D^2 = \theta(\theta - 1) \quad \dots(3)$$

Substituting equation (2) and (3) in equation (1), we get

$$(\theta(\theta - 1) - 3\theta + 4)y = e^{2z}$$

$$(\theta^2 - 4\theta + 4)y = e^{2z}$$

The Auxiliary equation is $m^2 - 4m + 4 = 0$

$$(m - 2)(m - 2) = 0$$

$$m = 2 \text{ (twice)}$$

$$\therefore \text{Complementary Function} = e^{2z}(Az + B)$$

$$= e^{2 \log x}(A \log x + B)$$

$$= e^{\log x^2}(A \log x + B)$$

$$\therefore \text{Complementary Function} = x^2(A \log x + B)$$

$$\begin{aligned} \text{Particular Integral} &= \frac{1}{\theta^2 - 4\theta + 4} e^{2z} = \frac{1}{4 - 8 + 4} e^{2z} \\ &= \frac{z}{2\theta - 4} e^{2z} \quad (\because \text{When we put } \theta = 2, \text{denominator is zero}) \\ &= \frac{z^2}{2} e^{2z} \quad (\because \text{When we put } \theta = 2, \text{denominator is zero}) \\ &= \frac{(\log x)^2}{2} e^{2 \log x} \\ &= \frac{x^2 (\log x)^2}{2} \end{aligned}$$

The Complete Solution is $y = \text{C.F.} + \text{P.I.}$

$$y = x^2(A \log x + B) + \frac{x^2(\log x)^2}{2}$$

Example 3: Solve $x^2 \frac{d^2 y}{dx^2} + 8x \frac{dy}{dx} + 13y = \log x$ **(Nov- 2005)**

Solution:

The given differential equation can be written as

$$(x^2 D^2 + 8xD + 13)y = \log x \quad \dots(1)$$

Put $x = e^z$ (or) $z = \log x$

$$xD = \theta \quad \dots(2)$$

$$x^2 D^2 = \theta(\theta - 1) \quad \dots(3)$$

Substituting equation (2) and (3) in equation (1), we get

$$(\theta(\theta - 1) + 8\theta + 13)y = z$$

$$(\theta^2 + 7\theta + 13)y = z$$

The auxiliary equation is $m^2 + 7m + 13 = 0$

$$m = \frac{-7 \pm \sqrt{49 - 4(13)}}{2} = \frac{-7 \pm \sqrt{49 - 52}}{2} = \frac{-7 \pm i\sqrt{3}}{2}$$

$$\therefore \text{Complementary Function} = e^{-\frac{7}{2}z} \left(A \cos \frac{\sqrt{3}}{2} z + B \sin \frac{\sqrt{3}}{2} z \right)$$

$$= e^{-\frac{7}{2} \log x} \left(A \cos \frac{\sqrt{3}}{2} \log x + B \sin \frac{\sqrt{3}}{2} \log x \right)$$

$$\therefore \text{Complementary Function} = x^{-\frac{7}{2}} \left(A \cos \frac{\sqrt{3}}{2} \log x + B \sin \frac{\sqrt{3}}{2} \log x \right)$$

$$\text{Particular Integral} = \frac{1}{\theta^2 + 7\theta + 13} z$$

$$\begin{aligned}
&= \frac{1}{13 \left[1 + \frac{7\theta + \theta^2}{13} \right]} z \\
&= \frac{1}{13} \left[1 + \frac{7\theta + \theta^2}{13} \right]^{-1} z \\
&= \frac{1}{13} \left(1 - \frac{7\theta}{13} \right) z \quad (\text{Neglecting the highest derivative since } \theta^2(z) = 0) \\
&= \frac{1}{13} \left(z - \frac{7}{13} \right) \\
&= \frac{13z - 7}{169} = \frac{13 \log x - 7}{169}
\end{aligned}$$

The Complete Solution is $y = \text{C.F.} + \text{P.I.}$

$$\therefore y = x^{-\frac{7}{2}} \left(A \cos \frac{\sqrt{3}}{2} \log x + B \sin \frac{\sqrt{3}}{2} \log x \right) + \frac{13 \log x - 7}{169}$$

Example 4: Solve $(x^2 D^2 + xD + 1)y = \log x$ **(Jan- 2010)**

Solution:

The given differential equation is $(x^2 D^2 + xD + 1)y = \log x$...(1)

Put $x = e^z$ (or) $z = \log x$

$$xD = \theta \quad \text{...(2)}$$

$$x^2 D^2 = \theta(\theta - 1) \quad \text{...(3)}$$

Substituting equation (2) and (3) in equation (1), we get

$$(\theta(\theta - 1) + \theta + 1)y = z$$

$$(\theta^2 + \theta - \theta + 1)y = z$$

$$(\theta^2 + 1)y = z$$

The auxiliary equation is $m^2 + 1 = 0$

$$m^2 = -1, m = \pm i$$

$\therefore$ Complementary Function = $A \cos z + B \sin z$

$$= A\cos(\log x) + B\sin(\log x)$$

$$\begin{aligned}\text{Particular Integral} &= \frac{1}{\theta^2 + 1} z \\ &= [1 + \theta^2]^{-1} z \\ &= [1 - \theta^2 + (\theta^2)^2 - \dots] z \\ &= z = \log x\end{aligned}$$

The Complete Solution is $y = \text{C.F.} + \text{P.I.}$

$$y = A\cos(\log x) + B\sin(\log x) + \log x$$

Example 5: Solve $x^3 \frac{d^3 y}{dx^3} + 9x^2 \frac{d^2 y}{dx^2} + 18x \frac{dy}{dx} + 6y = \frac{1}{x}$

Solution:

The given differential equation can be written as

$$(x^3 D^3 + 9x^2 D^2 + 18xD + 6)y = \frac{1}{x} \quad \dots(1)$$

Put $x = e^z$ (or) $z = \log x$

$$xD = \theta \quad \dots(2)$$

$$x^2 D^2 = \theta(\theta - 1) \quad \dots(3)$$

$$x^3 D^3 = \theta(\theta - 1)(\theta - 2) \quad \dots(4)$$

Substituting equation (2), (3) and (4) in equation (1), we get

$$(\theta(\theta - 1)(\theta - 2) + 9\theta(\theta - 1) + 18\theta + 6)y = \frac{1}{e^z}$$

$$(\theta^3 + 6\theta^2 + 11\theta + 6)y = e^{-z}$$

The auxiliary equation is $m^3 + 6m^2 + 11m + 6 = 0$

-1	1	6	11	6
	0	-1	-5	-6
	1	5	6	0

$\therefore -1$ is a root. The reduced equation is $m^2 + 5m + 6 = 0$

$$(m+2)(m+3) = 0$$

$$m_1 = -1, m_2 = -2, m_3 = -3$$

$\therefore$ Complementary Function = $Ae^{-z} + Be^{-2z} + Ce^{-3z}$

$$= Ae^{-\log x} + Be^{-2\log x} + Ce^{-3\log x}$$

$$= Ax^{-1} + Bx^{-2} + Cx^{-3}$$

$\therefore$ The Complete solution is $y = \frac{A}{x} + \frac{B}{x^2} + \frac{C}{x^3}$

$$\text{Particular Integral} = \frac{1}{\theta^3 + 6\theta^2 + 11\theta + 6} e^{-z}$$

$$= \frac{1}{-1 + 6 - 11 + 6} e^{-z} \quad (\because \text{Replacing } \theta \text{ by } -1)$$

$$= \frac{z}{3\theta^2 + 12\theta + 11} e^{-z} \quad (\because \theta = -1, \text{ denominator is zero})$$

$$= \frac{z}{3 - 12 + 11} e^{-z} \quad (\because \text{Replacing } \theta \text{ by } -1)$$

$$= \frac{z}{2} e^{-z} = \frac{\log x e^{-\log x}}{2}$$

$$= \frac{x^{-1} \log x}{2} = \frac{\log x}{2x}$$

The Complete Solution is $y = \text{C.F.} + \text{P.I.}$

$$\therefore y = \frac{A}{x} + \frac{B}{x^2} + \frac{C}{x^3} + \frac{\log x}{2x}$$

Example 6: Solve $(x^2 D^2 - 3xD - 5)y = \sin(\log x)$

(May- 2011)

Solution:

The given differential equation is $(x^2 D^2 - 3xD - 5)y = \sin(\log x) \quad \dots(1)$

Put $x = e^z$ (or) $z = \log x$

$$xD = \theta \quad \dots(2)$$

$$x^2 D^2 = \theta(\theta - 1) \quad \dots(3)$$

Substituting equation (2) and (3) in equation (1), we get

$$(\theta(\theta - 1) - 3\theta - 5)y = \sin z$$

$$(\theta^2 - 4\theta - 5)y = \sin z$$

The Auxiliary equation is $m^2 - 4m - 5 = 0$

$$(m + 1)(m - 5) = 0 \Rightarrow m_1 = -1, m_2 = 5$$

$$\begin{aligned} \therefore \text{Complementary Function} &= Ae^{-z} + Be^{5z} \\ &= Ae^{-\log x} + Be^{5\log x} \\ &= Ae^{\log x^{-1}} + Be^{\log x^5} \end{aligned}$$

$$\therefore \text{Complementary Function} = \frac{A}{x} + Bx^5$$

$$\begin{aligned} \text{Particular Integral} &= \frac{1}{\theta^2 - 4\theta - 5} \sin z \\ &= \frac{1}{-1 - 4\theta - 5} \sin z && (\because \text{Replacing } \theta^2 \text{ by } -1) \\ &= \frac{1}{-2(3 + 2\theta)} \frac{3 - 2\theta}{3 - 2\theta} \sin z \\ &= -\frac{1}{2} \frac{3 \sin z - 2\theta(\sin z)}{9 - 4\theta^2} \\ &= -\frac{1}{2} \frac{3 \sin z - 2 \cos z}{9 - 4(-1)} && (\because \text{Replacing } \theta^2 \text{ by } -1) \\ &= -\frac{1}{2} \frac{3 \sin(\log x) - 2 \cos(\log x)}{13} \end{aligned}$$

$$\therefore \text{Particular Integral} = \frac{2\cos(\log x) - 3\sin(\log x)}{26}$$

The Complete Solution is $y = \text{C.F.} + \text{P.I.}$

$$\therefore y = \frac{A}{x} + Bx^5 + \frac{2\cos(\log x) - 3\sin(\log x)}{26}$$

Example 7: Solve $x^2 y'' + 4xy' + 2y = 6x$ **(May-2007)**

Solution: The given differential equation can be written as

$$(x^2 D^2 + 4xD + 2)y = 6x \quad \dots(1)$$

Put $x = e^z$ (or) $z = \log x$

$$xD = \theta \quad \dots(2)$$

$$x^2 D^2 = \theta(\theta - 1) \quad \dots(3)$$

Substituting equation (2) and (3) in equation (1), we get

$$(\theta(\theta - 1) + 4\theta + 2)y = 6e^z$$

$$(\theta^2 + 3\theta + 2)y = 6e^z$$

The Auxiliary equation is $m^2 + 3m + 2 = 0$

$$(m + 1)(m + 2) = 0$$

$$m_1 = -1, m_2 = -2$$

$\therefore$ Complementary Function = $Ae^{-z} + Be^{-2z}$

$$= Ae^{-\log x} + Be^{-2\log x}$$

$$= Ae^{\log x^{-1}} + Be^{\log x^{-2}}$$

$\therefore$ Complementary Function = $\frac{A}{x} + \frac{B}{x^2}$

$$\text{Particular Integral} = \frac{1}{\theta^2 + 3\theta + 2} 6e^z$$

$$= \frac{1}{1 + 3 + 2} 6e^z \quad (\because \text{Replacing } \theta \text{ by } 1) \therefore$$

$$\text{Particular Integral} = e^z = e^{\log x} = x$$

The Complete Solution is $y = \text{C.F.} + \text{P.I.}$

$$\therefore y = \frac{A}{x} + \frac{B}{x^2} + x$$

Example 8: Solve $x^2 y'' + xy' + y = x + \log x$

(Nov-2004)

Solution:

The given differential equation can be written in the operator form as

$$(x^2 D^2 + xD + 1)y = x + \log x \quad \dots(1)$$

Put $x = e^z$ (or) $z = \log x$ and $e^{2z} = x^2$

$$xD = \theta \quad \dots(2)$$

$$x^2 D^2 = \theta(\theta - 1) \quad \dots(3)$$

Substituting equation (2) and (3) in equation (1), we get

$$\begin{aligned} (\theta(\theta - 1) + \theta + 1)y &= z + e^z \\ (\theta^2 + 1)y &= z + e^z \end{aligned}$$

The Auxiliary equation is $m^2 + 1 = 0$

$$m^2 = -1 \Rightarrow m = \pm i$$

$$\begin{aligned} \therefore \text{Complementary Function} &= A \cos z + B \sin z \\ &= A \cos(\log x) + B \sin(\log x) \end{aligned}$$

$$\therefore \text{Particular Integral} = \frac{1}{\theta^2 + 1} (z + e^z)$$

$$\begin{aligned} P.I_1 &= \frac{1}{\theta^2 + 1} (z) = [1 + \theta^2]^{-1} (z) = [1 - \theta^2 + \theta^4 - \dots](z) \\ &= z - 0 = z \end{aligned}$$

$$P.I_1 = \log x$$

$$P.I_2 = \frac{1}{\theta^2 + 1} (e^z) = \frac{1}{1 + 1} (e^z) = \frac{1}{2} e^z$$

$$P.I_2 = \frac{e^{\log x}}{2} = \frac{x}{2}$$

The Complete Solution is $y = \text{C.F.} + \text{P.I.}$

$$y = A \cos(\log x) + B \sin(\log x) + \log x + \frac{x}{2}$$

Example 9: Solve $x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} + y = x^2 \log x$ **(Nov- 2002)**

Solution:

The given differential equation can be written in the operator form as

$$(x^2 D^2 + xD + 1)y = x^2 \log x \quad \dots(1)$$

Put $x = e^z$ (or) $z = \log x$ and $e^{2z} = x^2$

$$xD = \theta \quad \dots(2)$$

$$x^2 D^2 = \theta(\theta - 1) \quad \dots(3)$$

Substituting equation (2) and (3) in equation (1), we get

$$(\theta(\theta - 1) + \theta + 1)y = ze^{2z}$$

$$(\theta^2 + 1)y = ze^{2z}$$

The Auxiliary equation is $m^2 + 1 = 0$

$$m^2 = -1$$

$$m = \pm i$$

$\therefore$ Complementary Function = $A \cos z + B \sin z$

$$= A \cos(\log x) + B \sin(\log x)$$

$$\text{Particular Integral} = \frac{1}{\theta^2 + 1} (ze^{2z})$$

$$P.I. = \frac{1}{\theta^2 + 1} (ze^{2z}) \quad (\because \text{Replacing } \theta \text{ by } \theta + 2)$$

$$= e^{2z} \frac{1}{\theta^2 + 4\theta + 5} (z)$$

$$= e^{2z} \frac{1}{5 \left[1 + \left(\frac{\theta^2 + 4\theta}{5} \right) \right]} (z)$$

$$\begin{aligned}
&= \frac{e^{2z}}{5} \left[1 + \left(\frac{\theta^2 + 4\theta}{5} \right) \right]^{-1} (z) \\
&= \frac{e^{2z}}{5} \left[1 - \left(\frac{\theta^2 + 4\theta}{5} \right) + \dots \right] (z) \\
&= \frac{e^{2z}}{5} \left[z - \frac{4(1)}{5} \right] \\
&= \frac{e^{2z}}{5} \left[\frac{5z - 4}{5} \right] \\
&= \frac{e^{2 \log x}}{5} \left[\frac{5 \log x - 4}{5} \right] = \frac{e^{\log x^2}}{5} \left[\frac{5 \log x - 4}{5} \right] \\
P.I. &= \left[\frac{x^2(5 \log x - 4)}{25} \right]
\end{aligned}$$

The Complete Solution is $y = C.F. + P.I.$

$$y = A \cos(\log x) + B \sin(\log x) + \frac{x^2(5 \log x - 4)}{25}$$

Example10: Solve $x^2 \frac{d^2 y}{dx^2} - 3x \frac{dy}{dx} + 4y = x^2 \cos(\log x)$ **(Apr-2003, 2008, Nov- 2008)**

Solution:

The given differential equation can be written in the operator form as

$$(x^2 D^2 - 3xD + 4)y = x^2 \cos(\log x) \quad \dots(1)$$

Put $x = e^z$ (or) $z = \log x$ and $e^{2z} = x^2$

$$xD = \theta \quad \dots(2)$$

$$x^2 D^2 = \theta(\theta - 1) \quad \dots(3)$$

Substituting equation (2) and (3) in equation (1), we get

$$(\theta(\theta - 1) - 3\theta + 4)y = e^{2z} \cos z$$

$$(\theta^2 - 4\theta + 4)y = e^{2z} \cos z$$

The auxiliary equation is $m^2 - 4m + 4 = 0$

$$(m - 2)^2 = 0$$

$$m = 2 \text{ (twice)}$$

$$\therefore \text{Complementary Function} = e^{2z} (Az + B)$$

$$= e^{2 \log x} (A \log x + B)$$

$$= x^2 (A \log x + B)$$

$$\therefore \text{Particular Integral} = \frac{1}{(\theta - 2)^2} (\cos z e^{2z})$$

$$P.I. = e^{2z} \frac{1}{(\theta + 2 - 2)^2} (\cos z) \quad (\because \text{Replacing } \theta \text{ by } \theta + 2)$$

$$= e^{2z} \frac{1}{(\theta)^2} (\cos z)$$

$$= -e^{2z} \cos z \quad (\because \text{Replacing } \theta^2 \text{ by } -1)$$

$$= -e^{\log x^2} \cos(\log x)$$

$$= -x^2 \cos(\log x)$$

The complete solution is $y = C.F + P.I$

$$\therefore y = x^2 [A \log x + B] - x^2 \cos(\log x)$$

Example 11: Solve $\frac{d^2 y}{dx^2} + \frac{1}{x} \frac{dy}{dx} = \frac{12 \log x}{x^2}$ **(Apr-2011)**

Solution: The given differential equation can be written as

$$x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} = 12 \log x$$

$$(x^2 D^2 + xD)y = 12 \log x \quad \dots(1)$$

Put $x = e^z$ (or) $z = \log x$

$$xD = \theta \quad \dots(2)$$

$$x^2 D^2 = \theta(\theta - 1) \quad \dots(3)$$

Substituting equation (2) and (3) in equation (1), we get

$$(\theta(\theta - 1) + \theta)y = 12z$$

$$\theta^2 y = 12z$$

The auxiliary equation is $m^2 = 0$

$$m = 0(\text{twice})$$

$$\therefore \text{Complementary Function} = e^{0z}(Az + B)$$

$$= A \log x + B$$

$$\text{Particular Integral} = \frac{1}{\theta^2}(12z)$$

$$P.I. = \frac{1}{\theta}(12 \frac{z^2}{2}) = 6 \frac{z^3}{3}$$

$$P.I. = 2(\log x)^3$$

The complete solution is $y = \text{C.F.} + \text{P.I.}$

$$y = A(\log x) + B + 2(\log x)^3$$

Example 12: Solve $x^2 \frac{d^2 y}{dx^2} + 4x \frac{dy}{dx} + 2y = x \log x$ **(Apr-2010)**

Solution: The given differential equation can be written in the operator form as

$$(x^2 D^2 + 4xD + 2)y = x \log x \quad \dots(1)$$

Put $x = e^z$ (or) $z = \log x$

$$xD = \theta \quad \dots(2)$$

$$x^2 D^2 = \theta(\theta - 1) \quad \dots(3)$$

Substituting equation (2) and (3) in equation (1), we get

$$(\theta(\theta-1)+4\theta+2)y = ze^z$$

$$(\theta^2 + 3\theta + 2)y = ze^z$$

The auxiliary equation is $m^2 + 3m + 2 = 0$

$$(m+1)(m+2) = 0 \Rightarrow m = -1, -2$$

$$\begin{aligned}\therefore \text{Complementary Function} &= Ae^{-z} + Be^{-2z} \\ &= Ae^{-\log x} + Be^{-2\log x} \\ &= \frac{A}{x} + \frac{B}{x^2}\end{aligned}$$

$$\text{Particular Integral} = \frac{1}{\theta^2 + 3\theta + 2}(ze^z)$$

$$\begin{aligned}P.I. &= e^z \frac{1}{(\theta+1)^2 + 3(\theta+1) + 2}(z) \quad (\because \text{Replacing } \theta \text{ by } \theta+1) \\ &= e^z \frac{1}{\theta^2 + 2\theta + 1 + 3\theta + 3 + 2}(z) = e^z \frac{1}{\theta^2 + 5\theta + 6}(z) \\ &= e^z \frac{1}{6 \left[1 + \left(\frac{\theta^2 + 5\theta}{6} \right) \right]}(z) \\ &= \frac{e^z}{6} \left[1 + \left(\frac{\theta^2 + 5\theta}{6} \right) \right]^{-1}(z) \\ &= \frac{e^z}{6} \left[1 - \left(\frac{\theta^2 + 5\theta}{6} \right) + \dots \right](z) \\ &= \frac{e^z}{6} \left[1 - \frac{5\theta}{6} \right](z) \\ &= \frac{e^z}{6} \left[z - \frac{5(1)}{6} \right] = \frac{e^z}{6} \left[\frac{6z - 5}{6} \right] \\ &= \frac{e^{\log x}}{6} \left[\frac{6 \log x - 5}{6} \right] \\ P.I. &= \left[\frac{x(6 \log x - 5)}{36} \right]\end{aligned}$$

The complete solution is $y = \text{C.F.} + \text{P.I.}$

$$\therefore y = \frac{A}{x} + \frac{B}{x^2} + \frac{x(6 \log x - 5)}{36}$$

Example 13: Solve $(x^2 D^2 + xD + 1)y = \sin(2 \log x) \sin(\log x)$ **(Apr- 2010)**

Solution: The given differential equation is

$$(x^2 D^2 + xD + 1)y = \sin(2 \log x) \sin(\log x) \quad \dots(1)$$

Put $x = e^z$ (or) $z = \log x$

$$xD = \theta \quad \dots(2)$$

$$x^2 D^2 = \theta(\theta - 1) \quad \dots(3)$$

Substituting equation (2) and (3) in equation (1), we get

$$(\theta(\theta - 1) + \theta + 1)y = \sin 2z \sin z$$

$$(\theta^2 + 1)y = \frac{1}{2}(\cos z - \cos 3z) \quad [\because 2 \sin A \sin B = \cos(A - B) - \cos(A + B)]$$

The auxiliary equation is $m^2 + 1 = 0$

$$m^2 = -1, m = \pm i$$

$\therefore$ Complementary Function = $A \cos z + B \sin z$

$$= A \cos(\log x) + B \sin(\log x)$$

$$\text{Particular Integral} = \frac{1}{2} \frac{1}{\theta^2 + 1} (\cos z - \cos 3z)$$

$$= \frac{1}{2} \left[\frac{1}{\theta^2 + 1} \cos z - \frac{1}{\theta^2 + 1} \cos 3z \right]$$

$$= P.I_1 + P.I_2$$

$$P.I_1 = \frac{1}{2} \left[\frac{1}{-1 + 1} \cos z \right]$$

Multiply by z and differentiate the denominator with respect to θ .

$$= \frac{z}{2} \left[\frac{1}{2\theta} \cos z \right] = \frac{z}{4} [\sin z]$$

$$P.I_2 = \frac{-1}{2} \left[\frac{1}{-9+1} \cos 3z \right] = \frac{-1}{2} \left[\frac{1}{-9+1} \cos 3z \right]$$

$$P.I_2 = \frac{1}{16} \cos 3z$$

$$P.I. = \frac{z \sin z}{4} + \frac{1}{16} \cos 3z$$

The complete solution is $y = C.F + P.I$

$$y = A \cos(\log x) + B \sin(\log x) + \frac{\log x \sin(\log x)}{4} + \frac{1}{16} \cos 3(\log x)$$

Example 14: Solve $(x^2 D^2 + xD + 1)y = \log x \sin(\log x)$

Solution:

The given differential equation is

$$(x^2 D^2 + xD + 1)y = \log x \sin(\log x) \quad \dots(1)$$

Put $x = e^z$ (or) $z = \log x$

$$xD = \theta \quad \dots(2)$$

$$x^2 D^2 = \theta(\theta - 1) \quad \dots(3)$$

Substituting equation (2) and (3) in equation (1), we get

$$(\theta(\theta - 1) + \theta + 1)y = z \sin z$$

$$(\theta^2 + 1)y = z \sin z$$

The Auxiliary equation is $m^2 + 1 = 0$

$$m^2 = -1, m = \pm i$$

$\therefore$ Complementary Function = $A \cos z + B \sin z$

$$= A \cos(\log x) + B \sin(\log x)$$

$$\text{Particular Integral} = \frac{1}{\theta^2 + 1} z \sin z$$

$$= \frac{1}{\theta^2 + 1} \text{Imaginary part of } e^{iz} z$$

$$\begin{aligned}
&= \text{Im.P.of } e^{iz} \frac{1}{(\theta+i)^2+1} z \quad (\because \text{Replacing } \theta \text{ by } \theta+1) \\
&= \text{Im.P.of } e^{iz} \frac{1}{2i\theta} \left[1 - \frac{i\theta}{2} \right]^{-1} z = \text{Im.P.of } -\frac{i}{2} e^{iz} \frac{1}{\theta} \left[1 + \frac{i\theta}{2} \right] z \\
&= \text{Im.P.of } -\frac{i}{2} e^{iz} \frac{1}{\theta} \left[z + \frac{i}{2} \right] = \text{Im.P.of } -\frac{i}{2} e^{iz} \frac{1}{\theta} \left[\frac{z^2}{2} + \frac{iz}{2} \right] \\
&= \text{Im.P.of } \frac{1}{2} (\sin z - i \cos z) \left[\frac{z^2}{2} + \frac{iz}{2} \right] = \frac{z}{4} \sin z - \frac{1}{4} z^2 \cos z
\end{aligned}$$

$$P.I = \frac{1}{4} \log x \sin(\log x) - \frac{1}{4} (\log x)^2 \cos(\log x)$$

$\therefore$ The Complete solution is $y = C.F + P.I$

$$y = A \cos(\log x) + B \sin(\log x) + \frac{1}{4} \log x \sin(\log x) - \frac{1}{4} (\log x)^2 \cos(\log x)$$

Exercise

Solve the following Differential Equations

1. $(x^2 D^2 + 3xD + 5)y = x \cos(\log x)$
2. $(x^3 D^3 + 3x^2 D^2 + xD + 1)y = \sin(\log x)$ (May- 2009)
3. $x^2 y'' - 3xy' + 4y = x^3 + x \log x$ (Nov-2006)
4. $x^2 \frac{d^2 y}{dx^2} - x \frac{dy}{dx} + y = x^2$ (Nov-2007)
5. $(x^2 D^2 - xD + 4)y = x^2 \sin(\log x)$
6. $(x^2 D^2 + 3xD)y = \frac{1}{(1-x)^2}$ (Nov-2011)
7. $(x^2 D^2 + 4xD + 2)y = e^x$

$$8. (x^3 D^3 + 3x^2 D^2 + xD - 8)y = 3x^2$$

$$9. (x^2 D^2 - xD + 1)y = \left(\frac{\log x}{x}\right)^2$$

$$10. (x^2 D^2 + 2xD - 20)y = (x+1)^2$$

Answers

$$1. y = x^{\frac{-1}{2}} A \cos(2 \log x) + B \sin(2 \log x) + \frac{x}{65} (4 \sin(\log x) + 7 \cos(\log x))$$

$$2. y = xA^{-1} + x^{\frac{1}{2}} \left[B \cos\left(\frac{\sqrt{3}}{2} \log x\right) + C \sin\left(\frac{\sqrt{3}}{2} \log x\right) \right] + \frac{1}{2} [\sin(\log x) + \cos(\log x)]$$

$$3. y = x^2 [A + B \log x] + x(\log x + 2)$$

$$4. y = x[A + B \log x] + x^2$$

$$5. y = x [A \cos \sqrt{3} \log x + B \sin \sqrt{3} \log x] + \frac{x^2}{13} [2 \cos(\log x) - 3 \sin(\log x)]$$

$$6. y = \frac{1}{x} (A + B \log x) + \frac{1}{x} \log \frac{x}{1-x}$$

$$7. y = \frac{A}{x} + \frac{B}{x^2} + \frac{e^x}{x^2}$$

$$8. y = Ax^2 + x^{-1} [B \cos \sqrt{3} \log x + C \sin \sqrt{3} \log x] + \frac{1}{4} x^2 \log x$$

$$9. y = (A \log x + B) \frac{1}{x} + \frac{1}{27x^2} [3(\log x)^2 + 4 \log x + 2]$$

$$10. y = \frac{A}{x^5} + Bx^4 - \left[\frac{1}{20} + \frac{x}{9} + \frac{x^2}{14} \right]$$

Simultaneous linear differential equations with constant coefficients

If x and y are two independent variables and t is the independent variable, then pair of equations of the form

$$f_1(D)x + f_2(D)y = T_1 \quad \dots(1)$$

$$\phi_1(D)x + \phi_2(D)y = T_2 \quad \dots(2)$$

is called a pair of simultaneous differential equations Where f_1, f_2, ϕ_1, ϕ_2 are polynomials in the operator $D = \frac{d}{dt}$ and T_1 and T_2 are functions of t . It is not possible to solve for the two dependent variables [unknowns], if only one of the above equation is given. If there are more than 2 independent variables, we should have as many equations as the number of dependent variables. To solve equations (1) and (2) simultaneously, we proceed as follows.

We operate both sides of (1) by $\phi_2 [D]$ and both sides of (2) by $f_2 [D]$ and subtract to eliminate y . Thus we get,

$$\begin{aligned} [f_1(D)\phi_2(D) - f_2(D)\phi_1(D)]x &= \phi_2(D)T_1 - f_2(D)T_2 \\ f(D)x &= T \end{aligned} \quad \dots(3)$$

Which is a linear equation in x and t with constant coefficients and can be solved by the known methods. The value of x obtained by the solution of (3) is substituted either in (1) or (2) to get the value of y . The number of arbitrary constants that appear in the values of x and y should be equal to the order of the resultant equation (3). If more arbitrary constants are introduced in the process of solving the equations, the extra ones should be expressed in terms of the other constants.

Example 1: Solve $\frac{dx}{dt} + 5x - 2y = 0; \frac{dy}{dt} + 2x + y = 0$

Solution:

The given equations are $\frac{dx}{dt} + 5x - 2y = 0$; $\frac{dy}{dt} + 2x + y = 0$

Denoting $\frac{d}{dt}$ by D, the given equations become,

$$Dx + 5x - 2y = 0 \quad \dots(1)$$

$$2x + (D+1)y = 0 \quad \dots(2)$$

$$\text{Multiplying (1) by 2} \Rightarrow 2(D+5)x - 4y = 0 \quad \dots(3)$$

$$\text{Operating (2) by } (D+5) \Rightarrow 2(D+5)x + (D+5)(D+1)y = 0 \quad \dots(4)$$

$$(3) - (4) \Rightarrow -4y - (D^2 + 6D + 5)y = 0$$

$$(D^2 + 6D + 9)y = 0$$

The auxiliary equation is $m^2 + 6m + 9 = 0$

$$(m+3)(m+3) = 0$$

$$m = -3 \text{ (twice)}$$

$$y = (At + B)e^{-3t}$$

From equation (2) we get, $2x = -(D+1)y = -[Dy + y]$

$$= -[D(At + B)e^{-3t} + (At + B)e^{-3t}]$$

$$= -[Ae^{-3t} + (At + B)e^{-3t}(-3) + (At + B)e^{-3t}]$$

$$= -[Ae^{-3t} - 2(At + B)e^{-3t}]$$

$$2x = 2(At + B)e^{-3t} - Ae^{-3t}$$

$$x = (At + B)e^{-3t} - \frac{A}{2}e^{-3t}$$

Example: 2 Solve $\frac{dx}{dt} - y = t$; $\frac{dy}{dt} + x = t^2$

Solution:

The given equations are $\frac{dx}{dt} - y = t$; $\frac{dy}{dt} + x = t^2$

Put $\frac{d}{dt} = D$ in the given equations, we get

$$Dx - y = t \quad \dots(1)$$

$$x + Dy = t^2 \quad \dots(2)$$

$$(2) \times D \Rightarrow Dx + D^2y = D(t^2) \quad \dots(3)$$

Solving (3) and (1), we get

$$(D^2 + 1)y = t$$

$\therefore$ The auxiliary equation is $m^2 + 1 = 0$

$$m^2 = -1$$

$$m = \pm i$$

$\therefore$ Complementary Function = $A \cos t + B \sin t$

$$\text{Particular Integral} = \frac{1}{D^2 + 1} t$$

$$= [1 + D^2]^{-1} t = (1 - D^2 + D^4 - \dots) t = t$$

$\therefore$ The complete solution is $y = \text{C.F} + \text{P.I}$

$$y = A \cos t + B \sin t + t$$

To find x from (2), we get

$$x + Dy = t^2$$

$$x = t^2 - Dy$$

$$= t^2 - D(A \cos t + B \sin t + t) \quad (\because \text{from (4)})$$

$$= t^2 - (-A \sin t + B \cos t + 1)$$

$$\therefore x = A \sin t - B \cos t + t^2 - 1$$

$$y = A \cos t + B \sin t + t$$

Example: 3 Solve $\frac{dx}{dt} + \frac{dy}{dt} + 2x + y = e^t; \frac{dy}{dt} + 3y + 5x = t$

(Nov-2002)

Solution:

The given equations are $\frac{dx}{dt} + \frac{dy}{dt} + 2x + y = e^t; \frac{dy}{dt} + 3y + 5x = t$

Put $\frac{d}{dt} = D$ in the given equations, we get (i.e.)

$$(i.e.) \quad Dx + Dy + 2x + y = e^t$$

$$(i.e.) \quad Dy + 3y + 5x = t$$

$$\therefore (D+2)x + (D+1)y = e^t \quad \dots(1)$$

$$5x + (D+3)y = t \quad \dots(2)$$

Here we eliminate x from (1) and (2)

$$(1) \times 5 \Rightarrow 5(D+2)x + 5(D+1)y = 5e^t \quad \dots(3)$$

$$(2) \times (D+2) \Rightarrow 5(D+2)x + (D+2)(D+3)y = (D+2)t$$

$$5(D+2)x + (D^2 + 5D + 6)y = 1 + 2t \quad \dots(4)$$

$$(4) - (3) \Rightarrow (D^2 + 5D + 6 - 5D - 6)y = 1 + 2t - 5e^t$$

$$(D^2 + 1)y = 1 + 2t - 5e^t$$

The auxiliary equation is $m^2 + 1 = 0$, $m^2 = -1$, $m = \pm i$

$\therefore$ Complementary Function = $A \cos t + B \sin t$

$$\text{Particular Integral} = \frac{1}{D^2 + 1} (1 + 2t - 5e^t)$$

$$P.I_2 = \frac{1}{D^2 + 1} (1 + 2t)$$

$$= [1 + D^2]^{-1} (1 + 2t) = [1 - D^2 + D^4 - \dots] (1 + 2t)$$

$$= 1 + 2t - 0 = 1 + 2t$$

$$P.I_2 = \frac{1}{D^2 + 1}(-5e^t) = \frac{1}{1+1}(-5e^t)$$

$$P.I_2 = -\frac{5}{2}e^t$$

∴ The complete solution is $y = C.F + P.I_1 + P.I_2$

$$y = A \cos t + B \sin t + (1+2t) - \frac{5}{2}e^t \quad \dots(5)$$

To find x from equation (2), we get $5x = t - (D+3)y$

$$= t - (D+3) \left(A \cos t + B \sin t + (1+2t) - \frac{5}{2}e^t \right) \quad (\because \text{from 5})$$

$$= t - D \left[A \cos t + B \sin t + (1+2t) - \frac{5}{2}e^t \right] - 3 \left[A \cos t + B \sin t + (1+2t) - \frac{5}{2}e^t \right]$$

$$= t - \left[-A \sin t + B \cos t + 2 - \frac{5}{2}e^t - 3A \cos t - 3B \sin t - 3 - 6t + \frac{15}{2}e^t \right]$$

$$= t + A \sin t - B \cos t - 2 + \frac{5}{2}e^t - 3A \cos t - 3B \sin t - 3 - 6t + \frac{15}{2}e^t$$

$$= t + (A-3B) \sin t - (3A+B) \cos t - 5 - 6t + 10e^t$$

$$\therefore x = \frac{1}{5} \left[t + (A-3B) \sin t - (3A+B) \cos t - 5 - 6t + 10e^t \right]$$

$$y = A \cos t + B \sin t + (1+2t) - \frac{5}{2}e^t$$

Example 4: Solve $\frac{dx}{dt} + y = \sin t$; $\frac{dy}{dt} + x = \cos t$, Given that $x = 2, y = 0$, when $t = 0$

(Nov-2007)

Solution:

The given equations can be written as

$$Dx + y = \sin t \quad \dots(1)$$

$$Dy + x = \cos t \quad \dots(2)$$

We eliminate x from (1) and (2),

$$(2) \times D \Rightarrow Dx + D^2 y = D(\cos t)$$

$$Dx + D^2 y = -\sin t \quad \dots(3)$$

$$(1) \Rightarrow Dx + y = \sin t$$

$$(3) - (1) \Rightarrow (D^2 - 1)y = -2\sin t$$

The auxiliary equation is $m^2 - 1 = 0$, $m^2 = 1$, $m = \pm 1$

Complementary Function = $Ae^t + Be^{-t}$

$$\text{Particular Integral} = \frac{1}{D^2 - 1}(-2\sin t)$$

$$= \frac{1}{-1 - 1}(-2\sin t) \quad (\text{Replacing } D^2 \text{ by } -1)$$

$$= \frac{2\sin t}{2} = \sin t$$

$\therefore$ The complete solution is $y = \text{C.F.} + \text{P.I.}$

$$y = Ae^t + Be^{-t} + \sin t \quad \dots(4)$$

To find x from equation (2),

$$x = \cos t - D(y)$$

$$x = \cos t - D[Ae^t + Be^{-t} + \sin t] \quad (\because \text{from 4})$$

$$= \cos t - Ae^t + Be^{-t} - \cos t$$

$$\therefore x = -Ae^t + Be^{-t}, \quad y = Ae^t + Be^{-t} + \sin t$$

$$\text{Given that } x = 2, y = 0$$

$$-A + B = 2 \quad \dots(5)$$

$$A + B = 0 \quad \dots(6)$$

Adding (5) and (6), we get $2B = 2$, $B = 1$

$$\therefore (6) \Rightarrow A + B = 0, A = -B, A = -1$$

$$\therefore x = e^t + e^{-t}$$

$$y = -e^t + e^{-t} + \sin t$$

Example 5: Solve $D^2x + y = \sin t, x + D^2y = \cos t$, where $D = \frac{d}{dt}$ **(Nov-2008)**

Solution:

The given equations are

$$D^2x + y = \sin t \quad \dots(1)$$

$$x + D^2y = \cos t \quad \dots(2)$$

We eliminate x ,

$$(2) \times D^2 \Rightarrow D^2x + D^4y = D^2(\cos t)$$

$$D^2x + D^4y = -\cos t \quad \dots(3)$$

$$(1) \Rightarrow D^2x + y = \sin t$$

$$(3) - (1) \Rightarrow (D^4 - 1)y = -(\sin t + \cos t)$$

The auxiliary equation is $m^4 - 1 = 0$

$$(m^2 + 1)(m^2 - 1) = 0$$

$$m^2 = -1, m^2 = 1$$

$$m = \pm i, m = \pm 1$$

Complementary Function = $A \cos t + B \sin t + ce^t + De^{-t}$

$$\text{Particular Integral} = \frac{1}{D^4 - 1}(-\sin t) + \frac{1}{D^4 - 1}(-\cos t)$$

$$= \frac{1}{1-1}(-\sin t) + \frac{1}{1-1}(-\cos t) \quad (\text{Replacing } D^4 \text{ by } 1)$$

$$= \frac{t}{4D^3}(-\sin t) + \frac{t}{4D^3}(-\cos t) \quad (\because Dr = 0)$$

$$= \frac{-t}{4D}(-\sin t) - \frac{t}{4D}(-\cos t) = \frac{t}{4}(-\cos t) - \frac{t}{4}(\sin t)$$

$$P.I. = \frac{-t \cos t}{4} - \frac{t \sin t}{4}$$

∴ The complete solution is $y = C.F + P.I$

$$y = A \cos t + B \sin t + ce^t + De^{-t} + \frac{-t \cos t}{4} + \frac{t \sin t}{4} \quad \dots(4)$$

To find x , from equation (2),

$$x + D^2 y = \cos t$$

$$x = \cos t - D^2 y \quad \dots(5)$$

$$D(y) = D \left[A \cos t + B \sin t + ce^t + De^{-t} + \frac{-t \cos t}{4} + \frac{t \sin t}{4} \right] \quad [\because \text{from 4}]$$

$$D^2 y = D(D(y))$$

$$= D \left[-A \sin t + B \cos t + ce^t - De^{-t} - \frac{\cos t}{4} + \frac{t \sin t}{4} + \frac{\sin t}{4} + \frac{t \cos t}{4} \right]$$

$$= -A \cos t - B \sin t + ce^t + De^{-t} + \frac{\sin t}{4} + \frac{\sin t}{4} + \frac{t \cos t}{4} + \frac{\cos t}{4} + \frac{\cos t}{4} - \frac{t \sin t}{4}$$

$$D^2(y) = -A \cos t - B \sin t + ce^t + De^{-t} + \frac{\sin t}{2} + \frac{\cos t}{2} + \frac{t}{4}(\cos t - \sin t)$$

From equation (5), $x = \cos t - D^2(y)$

$$\therefore x = \cos t + A \cos t + B \sin t - ce^t - De^{-t} - \frac{\sin t}{2} - \frac{\cos t}{2} + \frac{t}{4}(\sin t - \cos t)$$

$$y = A \cos t + B \sin t + ce^t + De^{-t} - \frac{t \cos t}{4} + \frac{t \sin t}{4}$$

Example 6: Solve $\frac{dx}{dt} + 2y + \sin t = 0, \frac{dy}{dt} - 2x - \cos t = 0$

Solution:

The given equations can be written as

$$Dx + 2y = -\sin t \quad \dots(1)$$

$$-2x + Dy = \cos t \quad \dots(2)$$

$$(1) \times 2 \Rightarrow 2Dx + 4y = -2\sin t \quad \dots(3)$$

$$(2) \times D \Rightarrow -2Dx + D^2y = D(\cos t)$$

$$-2Dx + D^2y = -\sin t \quad \dots(4)$$

$$2Dx + 4y = -2\sin t \quad \dots(3)$$

Adding (3) & (4) we get

$$(D^2 + 4)y = -3\sin t \quad \dots(5)$$

The auxiliary equation is $m^2 + 4 = 0$, $m = \pm 2i$

Complementary Function = $A\cos 2t + B\sin 2t$

$$\text{Particular Integral} = \frac{1}{D^2 + 4}(-3\sin t)$$

$$= \frac{-3}{-1 + 4}\sin t \quad \left[\text{Replacing } D^2 \text{ by } -1 \right]$$

$$= -\sin t$$

$$\therefore y = A\cos 2t + B\sin 2t - \sin t \quad \dots(6)$$

To find x , from equation (2),

$$2x = Dy - \cos t$$

$$= D[A\cos 2t + B\sin 2t - \sin t] - \cos t$$

$$= -2A\sin 2t + 2B\cos 2t - \cos t - \cos t$$

$$\therefore x = -A\sin 2t + B\cos 2t - \cos t$$

$$y = A\cos 2t + B\sin 2t - \sin t$$

Example 7: Solve $(D^2 - 5)x + 3y = \sin t$; $-3x + (D^2 + 5)y = t$ **(May-2012)**

Solution:

The given equations are

$$(D^2 - 5)x + 3y = \sin t \quad \dots(1)$$

$$-3x + (D^2 + 5)y = t \quad \dots(2)$$

To eliminate y,

$$(1) \times (D^2 + 5) \Rightarrow (D^4 - 25)x + 3(D^2 + 5)y = (D^2 + 5)\sin t$$

$$(i.e.) (D^4 - 25)x + 3(D^2 + 5)y = -\sin t + 5\sin t$$

$$(D^4 - 25)x + 3(D^2 + 5)y = 4\sin t \quad \dots(3)$$

$$(2) \times 3 \Rightarrow -9x + 3(D^2 + 5)y = 3t \quad \dots(4)$$

$$(3) - (4) \Rightarrow (D^4 - 16)x = 4\sin t - 3t$$

$\therefore$ The auxiliary equation is $m^4 - 16 = 0$, $m = \pm 2, \pm 2i$

Complementary Function = $Ae^{2t} + Be^{-2t} + c \cos 2t + D \sin 2t$

$$\text{Particular Integral} = \frac{1}{D^4 - 16}(4\sin t - 3t)$$

$$= 4 \cdot \frac{1}{D^4 - 16} \cdot \sin t - 3 \cdot \frac{1}{D^4 - 16} \cdot t$$

$$= 4 \frac{1}{1 - 16} \sin t - 3 \frac{1}{-16 \left(1 - \frac{D^4}{16} \right)} t$$

$$= -\frac{4}{15} \sin t + \frac{3}{16} \left[1 - \frac{D^4}{16} \right]^{-1} t$$

$$= -\frac{4}{15} \sin t + \frac{3}{16} \left(1 + \frac{D^4}{16} + \dots \right) t$$

$$= -\frac{4}{15} \sin t + \frac{3t}{16}$$

$$\therefore x = Ae^{2t} + Be^{-2t} + c \cos 2t + D \sin 2t - \frac{4}{15} \sin t + \frac{3t}{16} \quad \dots(5)$$

To find y, from equation (1),

$$3y = \sin t - (D^2 - 5)x = \sin t - D^2 x + 5x$$

$$D(x) = 2Ae^{2t} - 2Be^{-2t} - 2c \sin 2t + 2D \cos 2t - \frac{4}{15} \cos t$$

$$D^2(x) = 4Ae^{2t} + 4Be^{-2t} - 4c \cos 2t - 4D \sin 2t + \frac{4}{15} \sin t$$

$$\therefore 3y = \sin t - 4Ae^{2t} - 4Be^{-2t} + 4c \cos 2t + 4D \sin 2t - \frac{4}{15} \sin t$$

$$+ 5Ae^{2t} + 5Be^{-2t} + 5c \cos 2t + 5D \sin 2t - \frac{4}{3} \sin t + \frac{15t}{16}$$

$$= Ae^{2t} + Be^{-2t} + 9c \cos 2t + 9D \sin 2t - \frac{3}{5} \sin t + \frac{15}{16} t$$

$$\therefore y = \frac{A}{3} e^{2t} + \frac{B}{3} e^{-2t} + 3c \cos 2t + 3D \sin 2t - \frac{1}{5} \sin t + \frac{5}{16} t$$

Example 8: Solve $\frac{dx}{dt} + 2y = 5e^t$; $\frac{dy}{dt} - 2x = 5e^t$, given that $x = -1, y = 3$ when $t = 0$.

Solution:

(Nov-2006)

The given equations can be written as

$$Dx + 2y = 5e^t \quad \dots(1)$$

$$Dy - 2x = 5e^t \quad \dots(2)$$

To eliminate y,

$$(1) \times D \Rightarrow D^2 x + 2Dy = D(5e^t)$$

$$D^2 x + 2Dy = 5e^t \quad \dots(3)$$

$$(2) \times 2 \Rightarrow 2Dy - 4x = 10e^t \quad \dots(4)$$

$$(3)-(4) \Rightarrow (D^2 + 4)x = 5e^t - 10e^t = -5e^t$$

The auxiliary equation is $m^2 + 4 = 0$, $m^2 = -4$, $m = \pm 2i$

Complementary Function = $A \cos 2t + B \sin 2t$

$$\text{Particular Integral} = \frac{1}{D^2 + 4}(-5e^t) = \frac{1}{1+4}(-5e^t) \quad (\text{Replacing } D \text{ by } 1)$$

$$P.I = -e^t$$

$\therefore$ The complete solution is $x = C.F. + P.I.$

$$x = A \cos 2t + B \sin 2t - e^t \quad \dots(5)$$

To find y, from (1),

$$2y = 5e^t - Dx$$

$$2y = 5e^t - D[A \cos 2t + B \sin 2t - e^t]$$

$$= 5e^t + 2A \sin 2t - 2B \cos 2t + e^t$$

$$= 2A \sin 2t - 2B \cos 2t + 6e^t$$

$$\therefore y = A \sin 2t - B \cos 2t + 3e^t \quad \dots(6)$$

Given that when $x = -1, t = 0, (5) \Rightarrow A = 0$

When $y = 3, t = 0, (6) \Rightarrow B = 0$

$$\therefore x = -e^t$$

$$y = 3e^t$$

Exercise

Solve the following Differential equations

$$1. \quad \frac{dx}{dt} + 2x + 3y = 2e^{2t}; \frac{dy}{dt} + 3x + 2y = 0$$

$$2. \quad \frac{dx}{dt} + 2x - 3y = 5t; \frac{dy}{dt} - 3y + 2x = 2e^{2t} \quad (\text{Apr-2003})$$

3. $(D+2)x-3y=t; (D+2)y-3x=e^{2t}$ (Nov-2003)
4. $D^2y+(D+6)x=e^t; D^2y+Dx=t^2$
5. $\frac{dx}{dt}-y=t; \frac{dy}{dt}+x=t^2$
6. $\frac{dx}{dt}+2x-3y=t; \frac{dy}{dt}-3x+2y=e^{2t}$
7. $\frac{dx}{dt}-\frac{dy}{dt}+2y=\cos 2t; \frac{dx}{dt}+\frac{dy}{dt}-2x=\sin 2t$
8. $\frac{dx}{dt}-7x+y=0; \frac{dx}{dt}-2x-5y=0$

Answers

1. $x = Ae^t + Be^{-5t} + \frac{8}{7}e^{2t}; y = Ae^t + Be^{-5t} - \frac{6}{7}e^{2t}$
2. $x = Ae^t + Be^{-5t} + \frac{6}{7}e^{2t} - 2t - \frac{13}{5}; y = Ae^t - Be^{-5t} + \frac{8}{7}e^{2t} - 3t - \frac{12}{5}$
3. $x = -Ae^{-5t} + Be^t + \frac{3}{7}e^{2t} - \frac{2}{5}t - \frac{13}{25}; y = Ae^{-5t} + Be^t + \frac{4}{7}e^{2t} - \frac{3}{5}\left(t + \frac{4}{5}\right)$
4. $y = A + Be^{2t} + ce^{-2t} - \frac{1}{6}e^t + \frac{1}{3}t^2 + \frac{t}{3}$
 $x = -Be^{3t} - ce^{-2t} + \frac{1}{6}e^t - \frac{1}{3}t + \frac{1}{18}$
5. $x = A\cos t + B\sin t + t^2 - 1; y = -A\sin t + B\cos t + t$
6. $x = Ae^{-5t} + Be^t - \frac{2t}{5} - \frac{13}{25} + \frac{3e^{2t}}{7}$
 $y = -Ae^{-5t} + Be^t + ce^{-2t} + \frac{4}{7}e^{2t} - \frac{3t}{5} - \frac{12}{25}$

$$\begin{aligned}
 & x = e^t (A \cos t + B \sin t) - \frac{1}{2} \cos 2t \\
 7. & y = (e^t A \sin t - B \cos t) - \frac{1}{2} \sin 2t \\
 8. & x = e^{6t} (A \cos t + B \sin t); y = e^{2t} [(A - B) \cos t + (A + B) \sin t]
 \end{aligned}$$

Method of variation of parameters

This method is very useful in finding the general solution of the second order linear

$$\text{differential equation } \frac{d^2 y}{dx^2} + a_1 \frac{dy}{dx} + a_2 y = X \quad \dots(1)$$

where X is a function of x.

$$\text{Let the complementary function of (1) be C.F} = Au + Bv \quad \dots(2)$$

Where A,B are constants and u and v are function of x.

$$\text{Then P.I} = Pu + Qv, \quad \dots(3)$$

$$\text{Where } P = -\int \frac{vX}{uv' - u'v} dx \quad \dots(4)$$

$$Q = \int \frac{uX}{uv' - u'v} dx \quad \dots(5)$$

Substituting (4) and (5) in (3), we get the particular integral.

Hence the complete solution is $y = \text{C.F} + \text{P.I}$

$$y = Au + Bv + \text{P.I}$$

The solution of $\frac{d^2 y}{dx^2} + a_1 \frac{dy}{dx} + a_2 y = X$ is given by $y = \text{C.F} + \text{P.I}$

$$\text{C.F} = Au + Bv, \text{ P.I} = Pu + Qv$$

$$P = -\int \frac{vX}{uv' - u'v} dx, \quad Q = \int \frac{uX}{uv' - u'v} dx$$

Example 1: Solve $\frac{d^2 y}{dx^2} + a^2 y = \tan ax$ **using method of variation of parameters**

Solution:

The given equation can be written as $(D^2 + a^2)y = \tan ax$

The auxiliary equation is $m^2 + a^2 = 0$, $m^2 = -a^2$, $m = \pm ia$

Complementary Function = $A \cos ax + B \sin ax$

Here $u = \cos ax$; $v = \sin ax$

$$u' = -a \sin ax \quad v' = a \cos ax$$

$$uv' - vu' = \cos ax(a \cos ax) - \sin ax(-a \sin ax)$$

$$= a(\cos^2 ax + \sin^2 ax) = a$$

$$\begin{aligned} P &= -\int \frac{vX}{uv' - u'v} dx = -\int \frac{\sin ax \tan ax}{a} dx \\ &= -\frac{1}{a} \int \sin ax \cdot \frac{\sin ax}{\cos ax} dx \\ &= -\frac{1}{a} \int \frac{\sin^2 ax}{\cos ax} dx \\ &= -\frac{1}{a} \int \frac{1 - \cos 2ax}{\cos ax} dx \\ &= -\frac{1}{a} \int \left[\frac{1}{\cos ax} - \frac{\cos^2 ax}{\cos ax} \right] dx \\ &= -\frac{1}{a} \int (\sec ax - \cos ax) dx \end{aligned}$$

$$= -\frac{1}{a} \left[\frac{\log(\sec ax + \tan ax)}{a} - \frac{\sin ax}{a} \right]$$

$$P = \frac{1}{a^2} \sin ax - \frac{\log(\sec ax + \tan ax)}{a^2}$$

$$Pu = \frac{1}{a^2} \sin ax \cos ax - \frac{\cos ax \log(\sec ax + \tan ax)}{a^2}$$

$$\begin{aligned} Q &= \int \frac{ux}{uv' - vu'} dx = \int \frac{\cos ax \cdot \tan ax}{a} dx \\ &= \frac{1}{a} \int \cos ax \frac{\sin ax}{\cos ax} dx = \frac{1}{a} \int \sin ax dx \\ &= -\frac{\cos ax}{a^2} \end{aligned}$$

$$Qv = -\frac{\sin ax \cos ax}{a^2}$$

$$\therefore P.I = Pu + Qv$$

$$= -\frac{\cos ax \log(\sec ax + \tan ax)}{a^2}$$

$\therefore$ The complete solution is $y = C.F + P.I$

$$y = A \cos ax + B \sin ax - \frac{\cos ax \log(\sec ax + \tan ax)}{a^2}$$

Example 2: Solve $\frac{d^2 y}{dx^2} + 4y = 4 \tan 2x$ by using method of variation of

parameters.

(May-2009)

Solution:

The given equation can be written as $(D^2 + 4)y = 4 \tan 2x$

The auxiliary equation is $m^2 + 4 = 0$

$$m = \pm\sqrt{-4} = \pm 2i$$

$$\text{Complementary Function} = A\cos 2x + B\sin 2x$$

$$\text{Here } u = \cos 2x, \quad v = \sin 2x$$

$$u' = -2\sin 2x, \quad v' = 2\cos 2x$$

$$uv' - vu' = \cos 2x(2\cos 2x) - \sin 2x(-2\sin 2x) = 2$$

$$\begin{aligned} P &= -\int \frac{vX}{uv' - u'v} dx = -4 \int \frac{\sin 2x \tan 4x}{2} dx \\ &= -2 \int \sin 2x \frac{\sin 2x}{\cos 2x} dx = -2 \int \frac{\sin^2 2x}{\cos 2x} dx \\ &= -2 \int \frac{1 - \cos^2 2x}{\cos 2x} dx \\ &= -2 \int \frac{1}{\cos 2x} - \cos 2x dx \\ &= -2 \int (\sec 2x - \cos 2x) dx \\ &= -2 \left[\frac{\log(\sec 2x + \tan 2x)}{2} - \frac{\sin 2x}{2} \right] \end{aligned}$$

$$P = \sin 2x - \log(\sec 2x + \tan 2x)$$

$$Pu = \sin 2x \cos 2x - \cos 2x \log(\sec 2x + \tan 2x)$$

$$\begin{aligned} Q &= \int \frac{uX}{uv' - vu'} dx \\ &= \int \frac{\cos 2x 4 \tan 2x}{2} dx \\ &= 2 \int \cos 2x \frac{\sin 2x}{\cos 2x} dx = 2 \int \sin 2x dx \end{aligned}$$

$$= 2 \left(\frac{-\cos 2x}{2} \right) = -\cos 2x$$

$$Qv = -\sin 2x \cos 2x$$

$$\begin{aligned} \therefore PI = Pu + Qv &= \sin 2x \cos 2x - \cos 2x \log(\sec 2x + \tan 2x) - \sin 2x \cos 2x \\ &= -\cos 2x \log(\sec 2x + \tan 2x) \end{aligned}$$

The Complete Solution is $y = C.F. + P.I.$

$$\therefore y = A \cos 2x + B \sin 2x - \cos 2x \log(\sec 2x + \tan 2x)$$

Example 3: Solve $\frac{d^2 y}{dx^2} + y = \sec x$, by using method of variation of parameters.

(April-2003, Nov-2006, May-2008)

Solution:

The given equation can be written as $(D^2 + 1)y = \sec x$

The auxiliary equation is $m^2 + 1 = 0$

$$m^2 = -1$$

$$m = \pm i$$

Complementary Function = $A \cos 2x + B \sin 2x$

Here $u = \cos x, \quad v = \sin x$

$$u' = -\sin x \quad v' = \cos x$$

$$uv' - vu' = \cos^2 x + \sin^2 x = 1$$

$$P = -\int \frac{vX}{uv' - vu'} dx, \text{ where } X = \sec x$$

$$= -\int \sin x \sec x dx$$

$$= -\int \sin x \frac{1}{\cos x} dx = -\int \tan x dx$$

$$= \log \cos x$$

$$Pu = (\log \cos x) \cos x$$

$$\begin{aligned} Q &= \int \frac{uX}{uv' - vu'} dx \\ &= \int \cos x \sec x dx = \int dx = x \end{aligned}$$

$$Qv = x \sin x$$

$$P.I. = Pu + Qv = (\log \cos x) \cos x + x \sin x$$

The complete solution is $y = C.F. + P.I.$

$$\therefore y = A \cos x + B \sin x + (\log \cos x) \cos x + x \sin x$$

Example 4: Solve $y'' + y = \sec^2 x$ by using method of variation of parameters.

Solution:

The given equation can be written as $(D^2 + 1)y = \sec^2 x$

The auxiliary equation is $m^2 + 1 = 0$

$$m^2 = -1 \Rightarrow m = \pm i$$

$$\therefore C.F. = A \cos x + B \sin x$$

Here $u = \cos x, \quad v = \sin x$

$$u' = -\sin x \quad v' = \cos x$$

$$uv' - vu' = \cos^2 x + \sin^2 x = 1$$

$$\begin{aligned} P &= \int \frac{-vX}{uv' - vu'} dx, \text{ where } X = \sec^2 x \\ &= -\int \sin x \sec^2 x dx \end{aligned}$$

$$= -\int \frac{\sin x}{\cos^2 x} dx = -\int \sec x \cdot \tan x dx$$

$$P = -\sec x$$

$$Pu = -\sec x \cdot \cos x = -1$$

$$\begin{aligned} Q &= \int \frac{uX}{uv' - vu'} dx \\ &= \int \cos x \sec^2 x dx = \int \cos x \frac{\sec x}{\cos x} dx \\ &= \int \sec x dx \\ &= \log(\sec x + \tan x) \end{aligned}$$

$$Qv = \sin x \log(\sec x + \tan x)$$

$$\begin{aligned} P.I. &= Pu + Qv \\ &= -1 + \sin x \log(\sec x + \tan x) \end{aligned}$$

The complete solution is $y = C.F. + P.I.$

$$y = A \cos x + B \sin x - 1 + \sin x \log(\sec x + \tan x)$$

Example 5: Solve $\frac{d^2 y}{dx^2} + y = \operatorname{cosec} x$, **by using method of variation of parameters.**

(Nov-2004)

Solution:

The given equation can be written as $(D^2 + 1)y = 0$

The auxiliary equation is $m^2 + 1 = 0$, $m^2 = -1$, $m = \pm i$

Complementary Function = $A \cos x + B \sin x$

$$\text{Here } u = \cos x, \quad v = \sin x$$

$$u' = -\sin x \quad v' = \cos x$$

$$uv' - vu' = \cos^2 x + \sin^2 x = 1$$

$$P = \int \frac{-vX}{uv' - vu'} dx, \text{ where } X = \operatorname{cosec} x$$

$$= -\int \sin x \cos ecx dx = -\int dx = -x$$

$$Pu = -x \cos x$$

$$Q = \int \frac{uX}{uv' - vu'} dx$$

$$= \int \cos x \cos ecx dx = \int \cot x dx$$

$$= \log(\sin x)$$

$$Qv = \log(\sin x) \sin x$$

$$\text{Particular Integral} = -x \cos x + \log(\sin x) \sin x$$

The complete solution is $y = C.F. + P.I.$

$$y = A \cos x + B \sin x - x \cos x + \log(\sin x) \sin x$$

Example 6: Solve $(D^2 + 4D + 4)y = \frac{e^{-2}x}{x^2}$, using method of variation of

parameters.

(Nov-2002)

Solution:

The given equation is $(D^2 + 4D + 4)y = \frac{e^{-2}x}{x^2}$

The Auxiliary Equation is $m^2 + 4m + 4 = 0$

$$(m + 2)^2 = 0$$

$$\therefore m = -2 \text{ (twice)}$$

$$\therefore C.F. = e^{-2x} (Ax + B) = Axe^{-2x} + Be^{-2x}$$

Here $u = xe^{-2x}$, $v = e^{-2x}$

$$u' = e^{-2x} + xe^{-2x}(-2), v' = -2e^{-2x} = e^{-2x} - 2xe^{-2x}$$

$$\begin{aligned}
 uv' - vu' &= xe^{-2x}(-2e^{-2x}) - e^{-2x}(e^{-2x} - 2xe^{-2x}) \\
 &= -2e^{-4x} - e^{-4x} + 2xe^{-4x} \\
 &= -e^{-4x}
 \end{aligned}$$

$$\begin{aligned}
 P &= -\int \frac{vX}{uv' - vu'} dx, \text{ where } X = \frac{e^{-2x}}{x^2} \\
 &= -\int \frac{e^{-2x} \frac{e^{-2x}}{x^2}}{-e^{-4x}} dx = \int \frac{1}{x^2} dx = -\frac{1}{x}
 \end{aligned}$$

$$Pu = -\frac{1}{x} xe^{-2x} = -e^{-2x}$$

$$\begin{aligned}
 Q &= \int \frac{uX}{uv' - vu'} dx = \int \frac{xe^{-2x} \frac{e^{-2x}}{x^2}}{-e^{-4x}} dx = -\int \frac{dx}{x} \\
 &= -\log x
 \end{aligned}$$

$$Qv = -e^{-2x} \log x$$

$$P.I. = Pu + Qv = -e^{-2x}(1 + \log x)$$

The complete solution is $y = C.F. + P.I.$

$$y = (Ax + B)e^{-2x} - e^{-2x}(1 + \log x)$$

Example 7: Using the method of variation of parameter solve $\frac{d^2 y}{dx^2} + 4y = \tan 2x$

(Jan-2011, Jan-2013)

Solution:

The given equation can be written as $(D^2 + 4)y = \tan 2x$

The auxiliary equation is $m^2 + 4 = 0, \therefore m = \pm 2i$

Complementary Function = $A \cos 2x + B \sin 2x$

Here $u = \cos 2x$ $v = \sin 2x$

$$u' = -2\sin 2x \quad v' = 2\cos 2x$$

$$uv' - vu' = 2$$

$$P = -\int \frac{vX}{uv' - vu'} dx, \quad \text{where } X = \tan 2x$$

$$= -\int \frac{\sin 2x \tan 2x}{2} = -\frac{1}{2} \int \sin 2x \frac{\sin 2x}{\cos 2x} dx$$

$$= -\frac{1}{2} \int \frac{1 - \cos^2 2x}{\cos 2x}$$

$$= -\frac{1}{2} \int \frac{1}{\cos 2x} - \cos 2x dx = -\frac{1}{2} \int (\sec 2x - \cos 2x) dx$$

$$= -\frac{1}{2} \left[\frac{\log(\sec 2x + \tan 2x)}{2} - \frac{\sin 2x}{2} \right]$$

$$P = \frac{1}{4} [\sin 2x - \log(\sec 2x + \tan 2x)]$$

$$Pu = \frac{1}{4} [\sin 2x \cos 2x - \cos 2x \log(\sec 2x + \tan 2x)]$$

$$Q = \int \frac{uX}{uv' - vu'} dx = \int \frac{\cos 2x \tan 2x}{2} dx$$

$$= \frac{1}{2} \int \sin 2x dx = \frac{1}{2} \left(\frac{-\cos 2x}{2} \right) = \frac{-\cos 2x}{4}$$

$$Qv = -\frac{1}{4} \sin 2x \cos 2x$$

$$P.I. = Pu + Qv$$

$$= \frac{1}{4} [\sin 2x \cos 2x - \cos 2x \log(\sec 2x + \tan 2x) - \sin 2x \cos 2x]$$

$$= -\frac{1}{4} \cos 2x \log(\sec 2x + \tan 2x)$$

The complete solution is $y = C.F. + P.I.$

$$y = A \cos 2x + B \sin 2x - \frac{1}{4} \cos 2x \log(\sec 2x + \tan 2x)$$

Example 8: Solve $\frac{d^2 y}{dx^2} - y = \frac{2}{1+e^x}$ **by using method of variation of parameters.**

Solution:

The given equation can be written as $(D^2 - 1)y = \frac{2}{1+e^x}$

The Auxiliary Equation is $m^2 - 1 = 0$

$$m^2 = 1 \quad m = \pm 1$$

Complementary Function = $Ae^x + Be^{-x}$

$$\therefore C.F = Ae^x + Be^{-x}$$

$$\text{Here } u = e^x \quad v = e^{-x}$$

$$u' = e^x \quad v' = -e^{-x}$$

$$uv' - vu' = -e^x e^{-x} - e^{-x} e^x = -2$$

$$P = -\int \frac{vX}{uv' - vu'} dx, \text{ where } X = \frac{2}{1+e^x}$$

$$= -\int \frac{e^{-x} \left(\frac{2}{1+e^x} \right)}{-2} = \int \frac{e^{-x}}{1+e^x} dx$$

$$= \int \frac{e^{-2x}}{e^{-x} + 1} \quad [\because \text{Multiplied and divided by } e^{-x}]$$

$$\text{Put } t = e^{-x}$$

$$dt = -e^{-x} dx$$

$$dx = \frac{dt}{-e^{-x}} \quad (\text{i.e.}) \quad dx = -\frac{dt}{t}$$

$$\begin{aligned} \therefore P &= \int \frac{t^2}{1+t} \left(-\frac{dt}{t} \right) \\ &= \int \frac{t}{1+t} dt = -\int \frac{1+t-1}{1+t} dt \\ &= -\int \left(\frac{1+t}{1+t} - \frac{1}{1+t} \right) dt = -\int \left(1 - \frac{1}{1+t} \right) dt \\ &= -[t - \log(1+t)] \end{aligned}$$

$$P = \log(1+t) - t = \log(1+e^{-x}) - e^{-x}$$

$$Pu = e^x \log(1+e^{-x}) - 1$$

$$\begin{aligned} Q &= \int \frac{uX}{uv' - vu'} dx = \int \frac{e^x \left(\frac{2}{1+e^x} \right)}{-2} dx \\ &= \int \frac{e^x}{1+e^x} dx = -\log(1+e^x) \end{aligned}$$

$$Qv = -e^{-x} \log(1+e^x)$$

$$\therefore P.I. = Pu + Qv = e^x \log(1+e^{-x}) - 1 - e^{-x} \log(1+e^x)$$

The complete solution is $y = C.F + P.I$

$$\therefore y = Ae^x + Be^{-x} + e^x \log(1+e^{-x}) - 1 - e^{-x} \log(1+e^x)$$

Exercise

Solve the following equations using by the method of variation of parameters.

1. $(D^2 + 4)y = \sec 2x$

3. $y'' + y = \tan x$

2. $y'' - 2y' + 2y = e^x \tan x$

4. $\frac{d^2 y}{dx^2} + a^2 y = \sec ax$

$$5. \quad y'' - 4y' + 4y = \frac{e^{2x}}{x}$$

$$6. \quad y'' + y = x \sin x$$

$$7. \quad y'' - 2y' + 2y = e^x \tan x$$

Answers

$$1. \quad y = \left(A + \frac{1}{4} \log \cos 2x \right) \cos 2x + \left(B + \frac{x}{2} \sin 2x \right)$$

$$2. \quad y = e^x (A \cos x + B \sin x) - e^x \cos x \log(\sec x + \tan x)$$

$$3. \quad y = A \cos x + B \sin x - \cos x \log(\sec x + \tan x)$$

$$4. \quad y = \left(A + \frac{1}{a^2} \log \cos ax \right) \cos ax + \left(B + \frac{x}{a} \right) \sin ax$$

$$5. \quad y = (A - x)e^{2x} + (B + \log x)xe^{2x}$$

$$6. \quad y = A \cos x + B \sin x - x^2 \cos x + x \sin x$$

$$7. \quad y = e^x (c_1 \cos x + c_2 \sin x) - e^x \cos x \log(\sec x + \tan x)$$

PART A – QUESTIONS AND ANSWERS

1. Define Differential Equation.

Solution: A differential equation is an equation which involves differential co-efficient or derivatives.

Examples: (1) $e^x dx + e^y dy = 0$ (2) $\frac{d^2x}{dx^2} + n^2 x = 0$

2. Define Ordinary Differential Equation.

Solution: Differential equation which involves only one independent variable and the differential co-efficient are called ordinary differential equations.

Examples:

$$(i) e^x dx + e^y dy = 0$$

$$(ii) \frac{dx}{dy} - wy = a \cos pt$$

$$(iii) \frac{dy}{dt} + wx = a \sin pt$$

3. Write the general form of a linear differential equation of n^{th} order with constant co-efficient.

Solution:
$$a_0 \frac{d^n y}{dx^n} + a_1 \frac{d^{n-1} y}{dx^{n-1}} + a_2 \frac{d^{n-2} y}{dx^{n-2}} + \dots + a_n y = X$$

4. Define Order and Degree of differential equation.

Solution: The order of a differential equation is the order of the highest ordered derivative occurring in the differential equation. The degree of a differential equation is the degree of the highest derivative occurring in it. Then the fractional powers as far as the derivatives are concerned.

5. Define a linear differential equation.

Solution: Linear differential equations are those in which the dependent variable and its derivative occur only in the first degree and are not multiplied together.

6. Solve $\frac{d^2 y}{dx^2} - 5 \frac{dy}{dx} + 6y = 0$. (Jan -2011)

Solution: The given equation is $\frac{d^2 y}{dx^2} - 5 \frac{dy}{dx} + 6y = 0$

The equation can be written as $(D^2 - 5D + 6)y = 0$

The A.E Equation is $m^2 - 5m + 6 = 0$

(i.e.) $(m - 2)(m - 3) = 0$

$m_1 = 2 \quad m_2 = 3$

$\therefore \text{C.F} = Ae^{2x} + Be^{3x}$

7. Find the P.I of $(D^2 + 2D + 2)y = 5$

Solution: The P.I = $\frac{1}{D^2 + 2D + 2} 5 = \frac{1}{D^2 + 2D + 2} 5e^{0t}$ (replacing D by 0)

$$= \frac{1}{0 + 2(0) + 2} 5e^{0t}$$

$$\therefore \text{P.I.} = \frac{5}{2}$$

8. Find the P.I of $\frac{d^2y}{dx^2} + 4y = \sin 2x$ (Jan-2011)

Solution: Given that $(D^2 + 4)y = \sin 2x$

$$\begin{aligned} \text{P.I} &= \frac{\sin 2x}{f(x)} \\ &= \frac{\sin 2x}{D^2 + 4} \quad (\text{replacing } D^2 \text{ by } -4) \\ &= \frac{\sin 2x}{-4 + 4} = \frac{\sin 2x}{0} \end{aligned}$$

Multiplying the numerator by x and differentiating the denominator with respect to D.

$$\begin{aligned} &= x \frac{1}{2D} \sin 2x \\ \therefore \text{P.I.} &= \frac{-x \cos 2x}{4} \end{aligned}$$

9. Solve $\frac{d^4y}{dx^4} + 2n^2 \frac{d^2y}{dx^2} + n^4 y = 0$ (Apr- 2011)

Solution: The A.E is $m^4 + 2n^2 m^2 + n^4 = 0$

$$(m^2 + n^2)^2 = 0$$

$$(m^2 + n^2) = 0 \quad \text{twice}$$

$$m^2 = -n^2 \quad \text{twice}$$

$$m = \pm ni \text{ twice}$$

$$\therefore \text{C.F} = (A + Bx)\cos nx + (C + Dx)\sin nx$$

10. Find the P.I of $(D+1)^2 y = e^{-x}$ (Nov-2011)

$$\begin{aligned} \text{Solution: P.I} &= \frac{e^{-x}}{(D+1)^2} = \frac{e^{-x}}{(D^2 + 2D + 1)} = \frac{e^{-x}}{(1 - 2 + 1)} \quad (\text{replacing } D \text{ by } 2) \\ &= \frac{e^{-x}}{0} \end{aligned}$$

Multiplying the numerator by x and differentiating the denominator with respect to D .

$$\begin{aligned} &= \frac{xe^{-x}}{2D + 2} = \frac{xe^{-x}}{-2 + 2} \quad (\text{replacing } D \text{ by } 2) \\ \therefore \text{P.I} &= x \frac{e^{-x}}{0} \end{aligned}$$

Multiplying the numerator by x and differentiating the denominator with respect to D .

$$\therefore \text{P.I} = \frac{x^2 e^{-x}}{2}$$

11. Find the complementary function of $(D^2 - 3D + 2)y = e^{5x} + 2$ (Jan 2010)

$$\text{Solution: The A.E is } m^2 - 3m + 2 = 0$$

$$(m-1)(m-2) = 0$$

$$m_1 = 1, m_2 = 2$$

$$\therefore \text{C.F} = Ae^x + Be^{2x}$$

12. Find the Particular integral of $(D^2 - 3D + 4)y = e^{4x}$ (Jan-2010)

$$\text{Solution: P.I} = \frac{e^{4x}}{(D^2 - 3D + 4)} \quad (\text{replacing } D \text{ by } 4)$$

$$\text{P.I} = \frac{e^{4x}}{(16 - 12 + 4)} = \frac{e^{4x}}{8}$$

13. Solve $(D^2 - 4)y = e^{2x}$

(Jan-2013)

Solution: The Auxiliary equation is $m^2 - 4 = 0$

$$m^2 = 4$$

$$m = \pm 2$$

$$\therefore \text{C.F.} = Ae^{-2x} + Be^{2x}$$

$$\text{P.I.} = 0$$

$$\therefore \text{The complete solution is } y = Ae^{-2x} + Be^{2x}$$

14. Solve $(D^4 - 1)y = 0$

Solution: The auxiliary equation is $(m^4 - 1) = 0$

$$(m^2 - 1)(m^2 + 1) = 0$$

$$(m - 1)(m + 1)(m^2 + 1) = 0$$

$$m = 1, -1, \pm i$$

$$\therefore \text{The general solution } y = Ae^{-x} + Be^x + C \cos x + D \sin x$$

15. Find the complementary function of $(D^2 + 3D + 4)y = e^{2x}$

Solution: The auxiliary equation is $m^2 + 3m + 4 = 0$

$$m = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$m = \frac{-3 \pm \sqrt{9 - 16}}{2} = \frac{-3 \pm \sqrt{-7}}{2} = \frac{-3 \pm i\sqrt{7}}{2}$$

$$\therefore \text{C.F.} = e^{-\frac{3}{2}x} \left(A \cos \frac{\sqrt{7}}{2} x + B \sin \frac{\sqrt{7}}{2} x \right)$$

16. Find the complementary function of $(D^2 + 2D + 1)y = \cos 2x$

(Apr 2010)

Solution: The A.E. is $m^2 + 2m + 1 = 0$

$$(m+1)(m+1)=0$$

$$m_1 = -1, \quad m_2 = -1$$

$$\therefore \text{C.F.} = (Ax + B)e^{-x}$$

17. Find the P.I of $(D^2 + 16)y = \cos 4x$

(Apr - 2010)

Solution: P.I = $\frac{\cos 4x}{(D^2 + 16)}$ (replacing D^2 by -4)

$$= \frac{\cos 4x}{(-16 + 16)} = \frac{\cos 4x}{0}$$

Multiplying the numerator by x and differentiating the denominator with respect to D.

$$\text{P.I} = \frac{x \cos 4x}{2D} = \frac{x}{2} \int \cos 4x \, dx$$

$$\therefore \text{P.I.} = \frac{x \sin 4x}{8}$$

18. Find the C.F of $(D^2 - 4D + 4)y = e^{2x} + \cos 2x$

(Apr - 2012)

Solution: The A.E is $m^2 - 4m + 4 = 0$

$$(m-2)(m-2) = 0$$

$$m = 2, \quad m = 2$$

$$\therefore \text{C.F} = (Ax + B)e^{2x}$$

19. Find the C.F of $(D^2 - 4D - 12)y = \sin x \sin 2x$

(Nov- 2011)

Solution: The A.E is $m^2 - 4m - 12 = 0$

$$(m+2)(m-6) = 0$$

$$m_1 = -2 \quad m_2 = 6$$

$$\therefore \text{C.F} = Ae^{-2x} + Be^{6x}$$

20. Find the P.I of $(D^2 + 4)y = x^2$

(Nov -2011)

Solution: P.I = $\frac{x^2}{D^2 + 4}$

$$= \frac{x^2}{4\left(1 + \frac{D^2}{4}\right)} = \frac{1}{4}\left(1 + \frac{D^2}{4}\right)^{-1} x^2$$

[We know that $(1+x)^{-1} = 1 - x + x^2 - x^3 + \dots$]

$$= \frac{1}{4}\left(1 - \frac{D^2}{4} + \left(\frac{D^2}{4}\right)^2\right)x^2 = \frac{x^2}{4} - \frac{D^2}{4}\left(\frac{x^2}{4}\right)$$

$$= \frac{x^2}{4} - \frac{D}{4}D\left(\frac{x^2}{4}\right) = \frac{x^2}{4} - \frac{D}{4}\left(\frac{2x}{4}\right)$$

$$= \frac{x^2}{4} - \frac{1}{4}\left(\frac{2}{4}\right) = \frac{x^2}{4} - \frac{2}{16}$$

$$\therefore \text{P.I} = \frac{x^2}{4} - \frac{1}{8}$$

21. Find the complementary function of $(D^3 - D^2 - 6D)y = x^2 + 1$ (Nov -2011)

Solution: The A. E is $m^3 - m^2 - 6m = 0$

$$m(m^2 - m - 6) = 0$$

$$m = 0 \quad (\text{or}) \quad (m^2 - m - 6) = 0$$

$$(m-3)(m+2) = 0$$

$$m_1 = 0, m_2 = 3, m_3 = -2$$

$$\therefore \text{C. F. Y} = A + Be^{3x} + Ce^{-2x}$$

22. Find the particular integral of $(D-1)^2 y = e^x \sin x$

Solution: The given equation is $(D-1)^2 y = e^x \sin x$

$$\begin{aligned}
\text{Particular Integral} &= \frac{1}{(D-1)^2} e^x \sin x \\
&= e^x \frac{1}{(D+1-1)^2} \sin x && \text{(Replacing D by D+1)} \\
&= e^x \frac{1}{D^2} \sin x = e^x \frac{1}{D} \int -\cos x dx = e^x (-\sin x) \\
\therefore \text{P.I.} &= -e^x \sin x
\end{aligned}$$

23. Find the particular integral of $(D^2 + 4D + 4)y = xe^{-2x}$

Solution: The given equation is $(D^2 + 4D + 4)y = xe^{-2x}$

$$\begin{aligned}
\text{Particular Integral} &= \frac{1}{D^2 + 4D + 4} xe^{-2x} \\
&= e^{-2x} \frac{1}{(D-2)^2 + 4(D-2) + 4} x && \text{(Replacing D by (D-2))} \\
&= e^{-2x} \frac{1}{D^2 + 4 - 4D + 4D - 8 + 4} x \\
&= e^{-2x} \frac{1}{D^2} x = e^{-2x} \frac{1}{D} \int x dx \\
&= e^{-2x} \int \frac{x^2}{2} dx \\
\therefore \text{P.I.} &= e^{-2x} \frac{x^3}{6}
\end{aligned}$$

24. Solve $x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} + 2y = 0$ **(Jan-2013)**

Solution: The given differential equation can be written as $(x^2 D^2 + xD + 2)y = 0$

$$\text{Put } xD = \theta: x^2 D^2 = \theta(\theta-1)$$

$$\therefore \theta(\theta-1) + \theta + 2 = 0$$

$$\theta^2 + 2 = 0$$

The auxiliary equation is $m^2 + 2 = 0$

$$m^2 = -2$$

$$m = \pm i\sqrt{2}$$

$$\therefore \text{C.F.} = A \cos \sqrt{2}z + B \sin \sqrt{2}z \quad \text{P.I.} = 0$$

$\therefore$ The complete solution is $y = A \cos \sqrt{2} \log x + B \sin \sqrt{2} \log x$

25. Solve $x^2 y'' - 20y = 0$

Solution: The equation is $(x^2 D^2 - 20)y = 0$... (1)

Putting $x = e^z$, $Z = \log x$

$$xD = \theta \quad \dots (2)$$

$$x^2 D^2 = \theta(\theta - 1) \quad \dots (3)$$

Substituting (2) and (3) in (1) we get $(\theta^2 - \theta - 20)y = 0$

The auxiliary equation is $(m^2 - m - 20) = 0$

$$(m - 5)(m + 4) = 0$$

$$m = -4, 5$$

$$y = Ae^{-4Z} + Be^{5Z}$$

$$= Ae^{-4 \log x} + Be^{5 \log x}$$

$$= Ae^{\log x^{-4}} + Be^{\log x^5}$$

$$\therefore y = Ax^{-4} + Bx^5$$

26. How do you solve a second order differential equation by the method of variation of parameters?

(Apr - 2012)

Solution: The general form of the second order differential equation is

$$\frac{d^2y}{dx^2} + a_1 \frac{dy}{dx} + a_2 y = X \text{ Where } X \text{ is a function of } x \text{ (or) constant}$$

Let the complementary function of this equation be

$$\text{C.F} = Af_1 + Bf_2 \quad \dots(1)$$

Where A and B are arbitrary constants and f_1 and f_2 are function of x Then

$$\text{P.I} = Pf_1 + Qf_2 \quad \dots(2)$$

$$\text{where } P = - \int \frac{f_2 X}{f_1 f_2' - f_2' f_1} dx \quad \dots(3)$$

$$Q = \int \frac{f_1 X}{f_1 f_2' - f_2' f_1} dx \quad \dots(4)$$

Substituting (3) and (4) in (2)

The general solution is $y = \text{C.F} + \text{P.I}$

$$\text{(i.e.) } y = Af_1 + Bf_2 + P.I$$

27. Define electric circuits.

Solution: An electric circuit consists of a resistor and an inductor or capacitor in series with a source of electromotive force

R(ohms) - Resistance

L - Inductance

C - Capacitance

E - Electromotive force

28. Write the three types of electric circuits as being built of elements.

Solution:

- (i) **Resistance:** Resistance is that circuit parameter which opposes the flow of current $e = iR$ (i.e.) The voltage drop across a resistance R.
- (ii) **Inductance:** Inductance is that circuit parameter which opposes a change in current.

$e = L \frac{di}{dt}$ (i.e.) the voltage drop across an inductance L is the rate of change of current in times.

(iii) **Capacitance:** Capacitance is that circuit parameter which opposes a change in

voltage $e = \frac{q}{c}$, since the current is the rate of flow of positive charge $i = \frac{dq}{dt}$

29. Write two Kirchhoff's law.

Solution: In a closed circuit, the sum of the voltage drops across each element of the Circuit is equal to the impressed voltage. At a junction of two or more elements of a circuit, the amount of current into the junction point is equal to the amount of current away from the junction point. $L \frac{di}{dt} + Ri = E$ Where i is the current if E is constant (or) a function of (t)

30. Solve $L \frac{d^2q}{dt^2} + R \frac{dq}{dt} + \frac{1}{C}q = 0$ **given** $R^2 = \frac{4L}{C}$

Solution: The given equation is $\left(LD^2 + RD + \frac{1}{C} \right)q = 0$ where $D = \frac{d}{dt}$

The auxiliary equation is $Lm^2 + Rm + \frac{1}{C} = 0$

$$m = \frac{-R \pm \sqrt{R^2 - \frac{4L}{C}}}{2L} = \frac{-R \pm 0}{2L} \quad \left[\because R^2 = \frac{4L}{C} \right]$$

Hence $q = e^{-\frac{Rt}{2L}}(A + Bt)$.